Disclaimer: Slide material is provided as framework. Slides do NOT substitute online material, meeting attendance, and further reading.

Digital Health [DiHe]

Introduction

Prof. Dr. Oliver Amft

Intelligent Embedded Systems Lab
e-mail: amft@computer.org

Before we start...

1) Lectures & exercises sessions: In presence

2) Materials repository: ILIAS

3) Participation is required

Please connect to ILIAS

Link:
https://ilias.uni-freiburg.de/goto.php/grp/4384010

Password: digitalhealth2627

Deadline: 29.10.2026, 23:59

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Topics and learning objectives

1. Digital Health: Overview
2. Medical data integration approach
3. Context awareness: Definition
4. Context awareness: Theory & realisation
5. Digital Therapeutics (DTx) overview

Please connect to ILIAS

Link:
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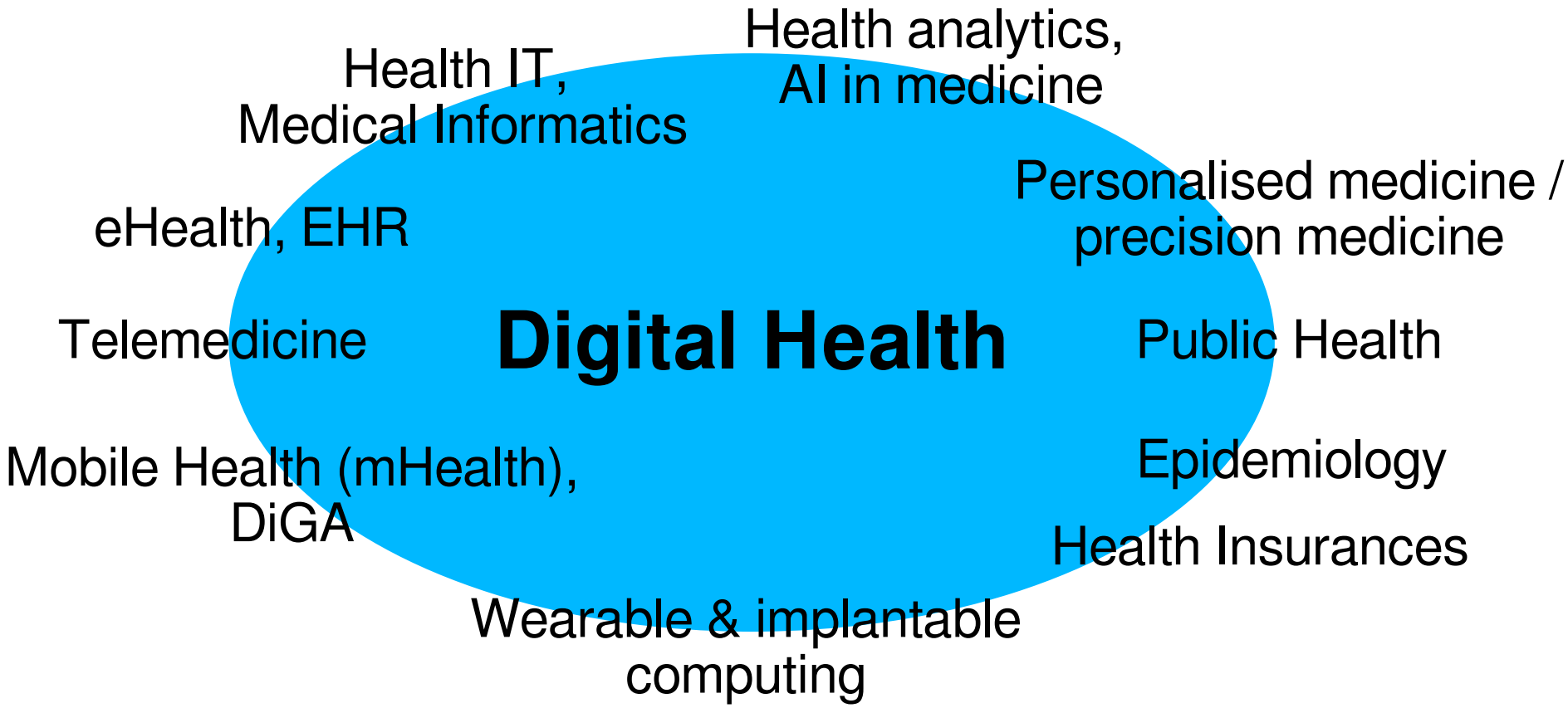
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Digital Health: Overview

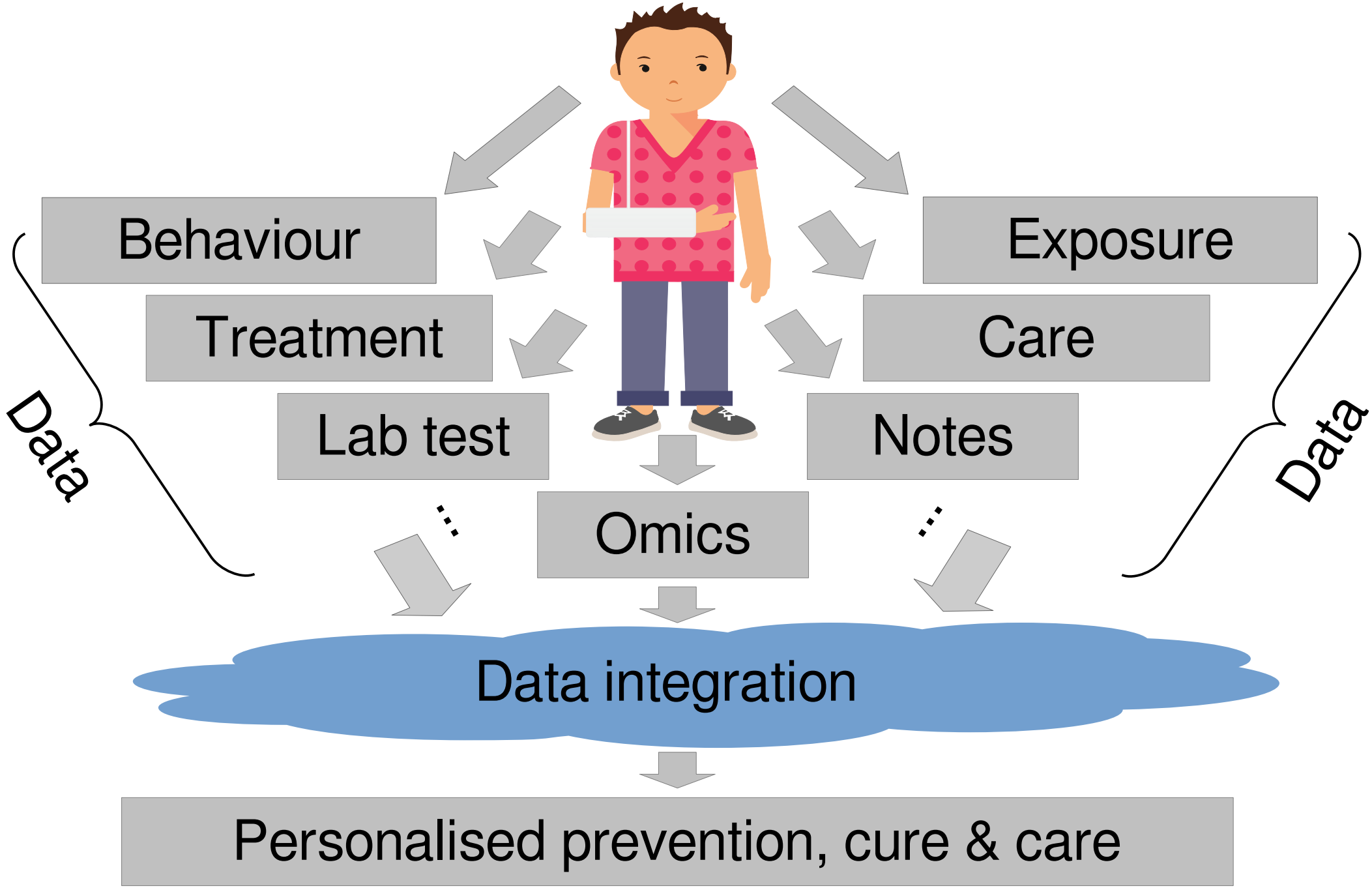
Prof. Dr. Oliver Amft

amft@computer.org

Related terms & areas



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Amft, Perv Comput 17(1), 2018

Medical data summary

- **Medical datum:** any single observation of a patient, defined by 4 elements:
  1) Patient in question,
  2) parameter being observed,
  3) value of the parameter,
  4) time of observation.

  The datum may have an uncertainty associated to it; sampling strategies apply.

- **Medical data:** collection of observations.

- **Types of medical data:**
  - Narrative, text, coding: diagnosis, conditions, etc.
  - Numerical measurements: lab tests, e.g. blood test, vital signs.
  - Recorded signals: timeseries, e.g. ECG.
  - Images: e.g. radiology images, sketches.

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Why integrate data?

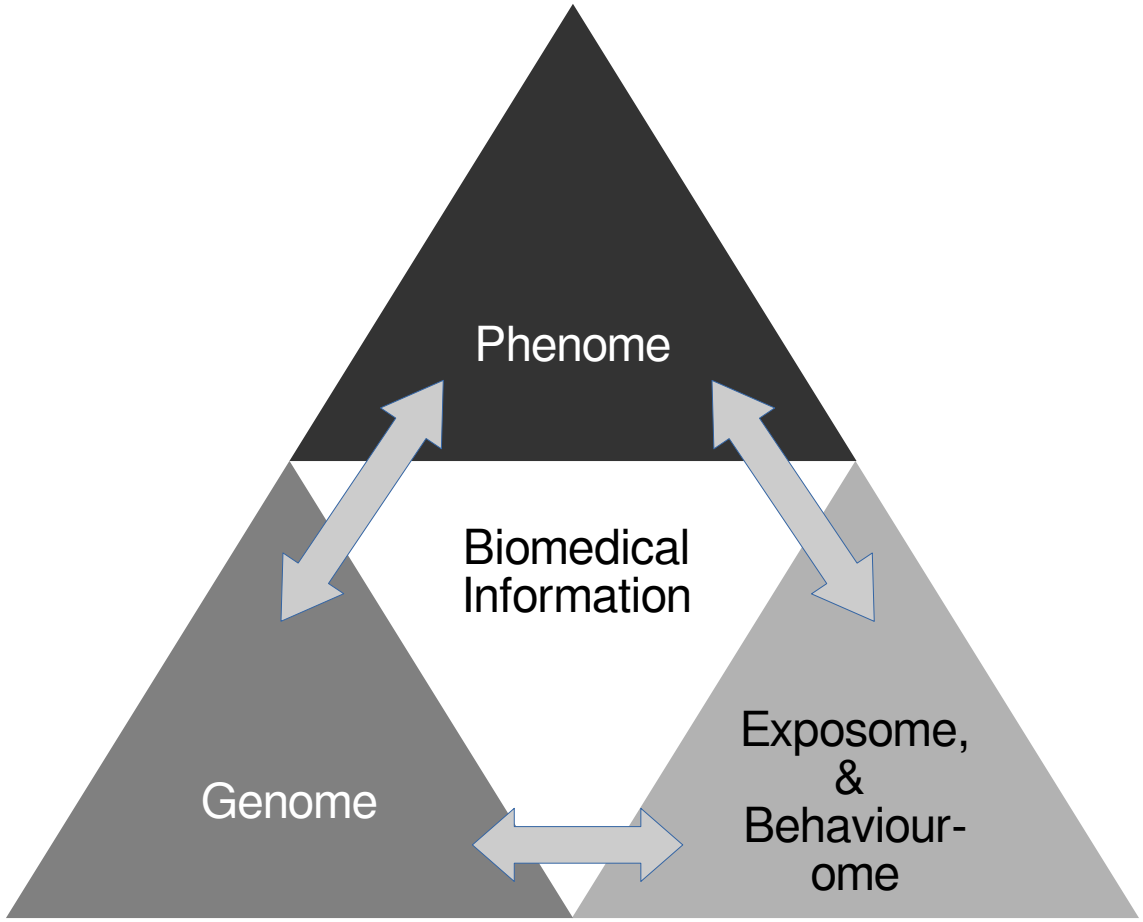


Answer typical questions:

- *What is the best evidence-based practice to treat disease D under consideration of patient age and co-condition C ?*
- *What condition/disease have symptoms X, Y, Z been attributed to?*
- *What happened to patients like me, at your institution, who chose treatment A vs. treatment B ?*
- *What are the treatment costs for disease D ?*

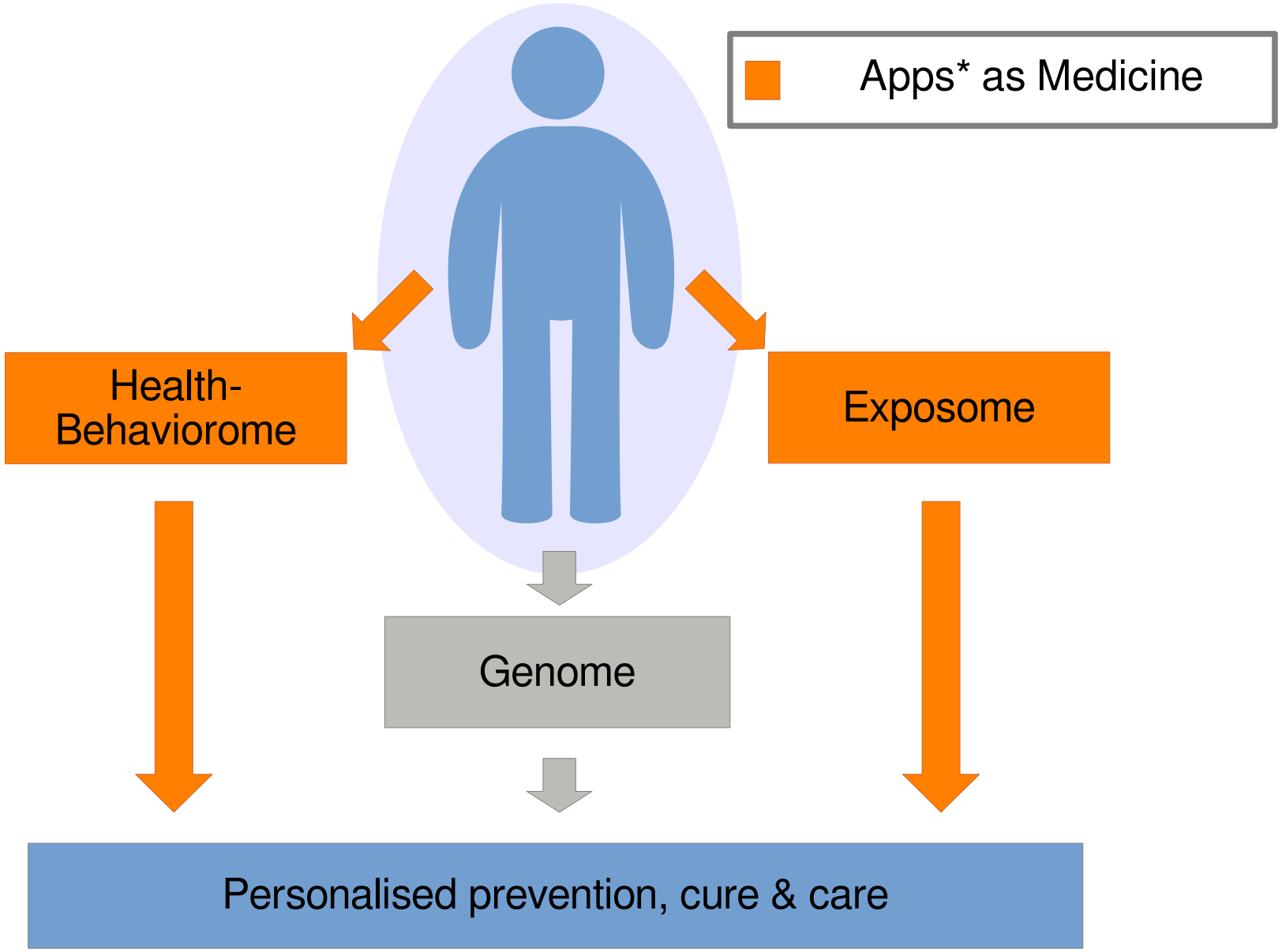
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Genome x Exposome



Sanchez et al., J Am Med Inform Assoc, 2014
Amft, Perv Comput, 2018

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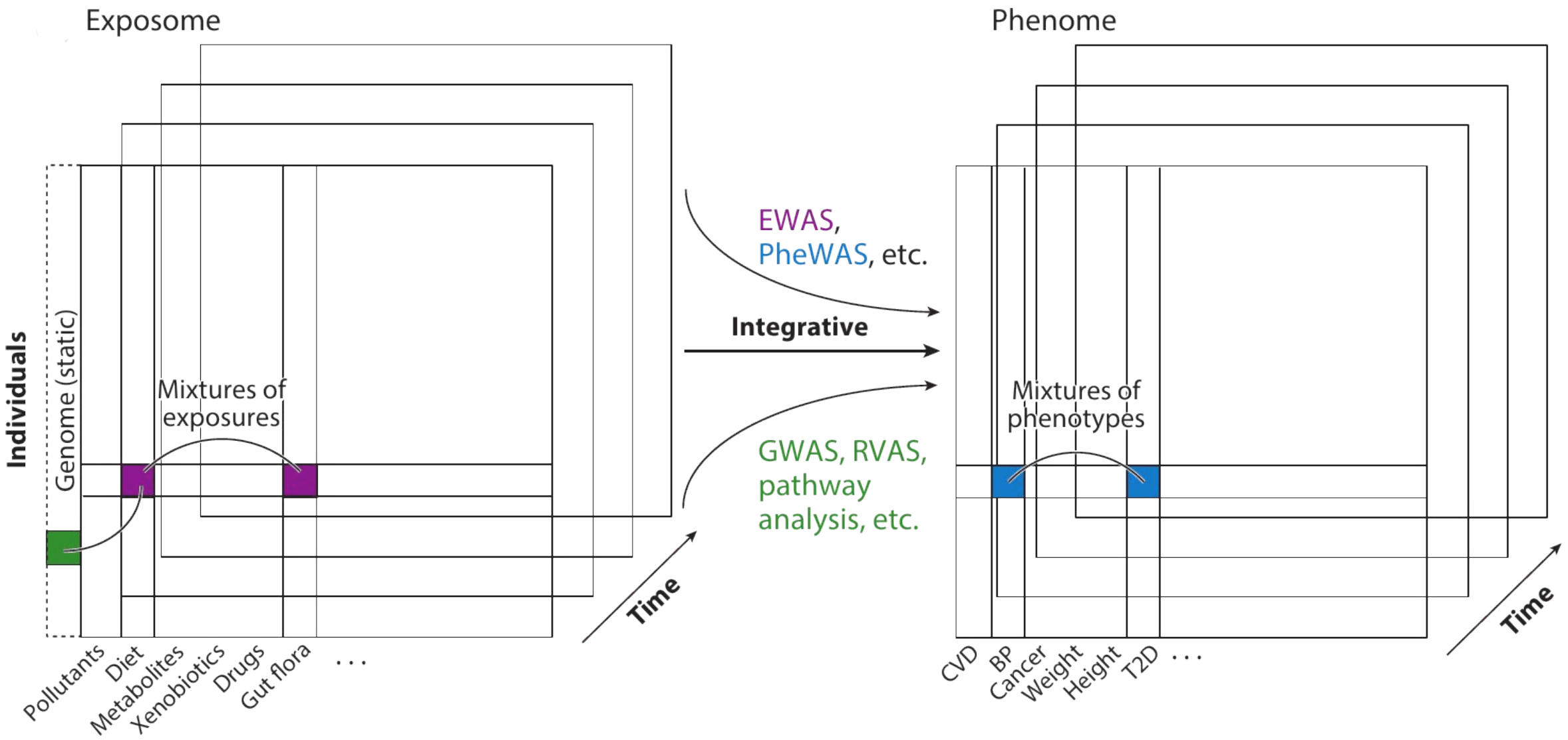


* Can include wearables, implants...

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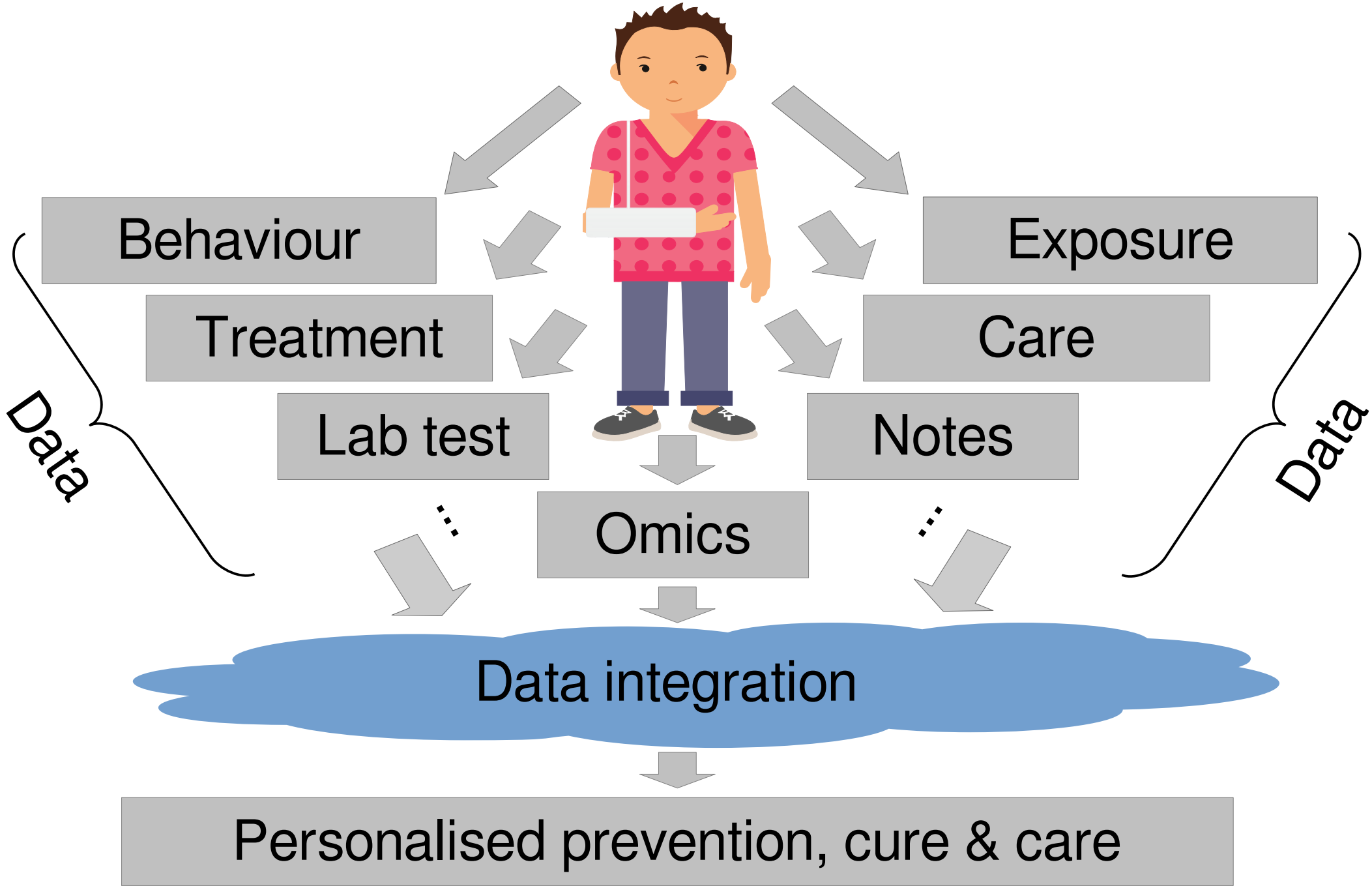
Amft, Perv Comput, 2018

Genome x Exposome = Phenome



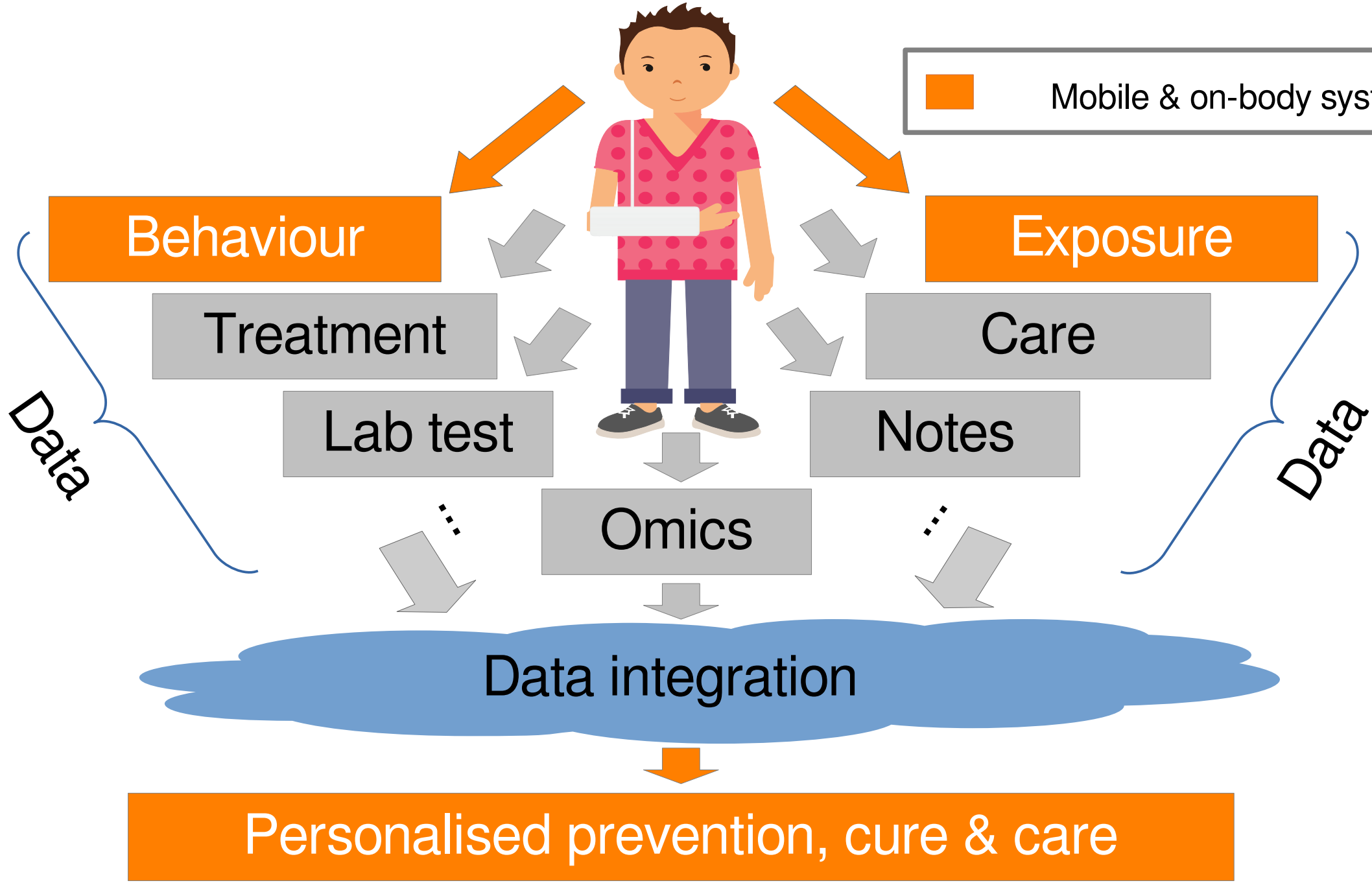
Manrai et al., 2017

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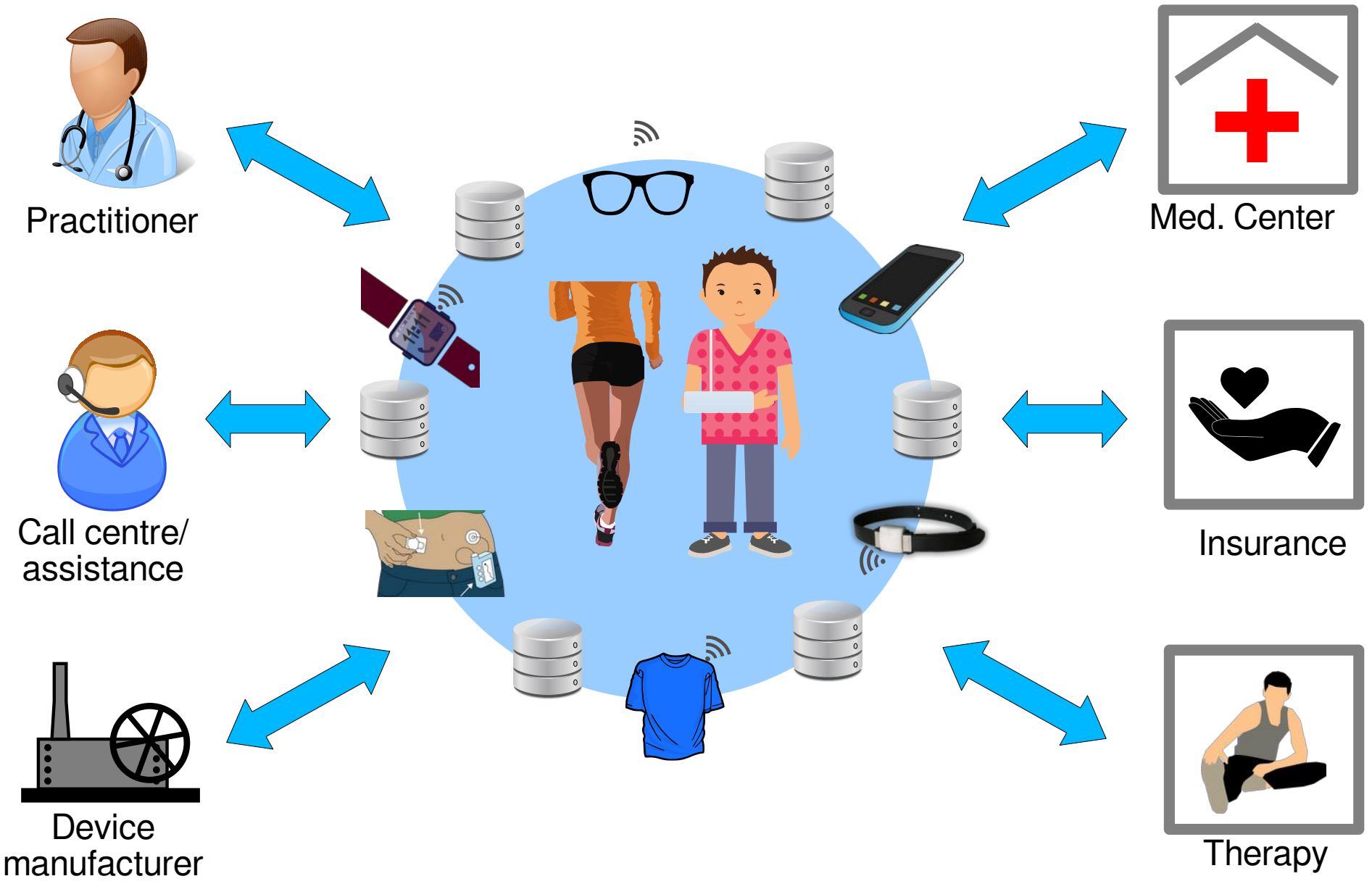
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Amft, Perv Comput, 2018



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Amft, Perv Comput, 2018



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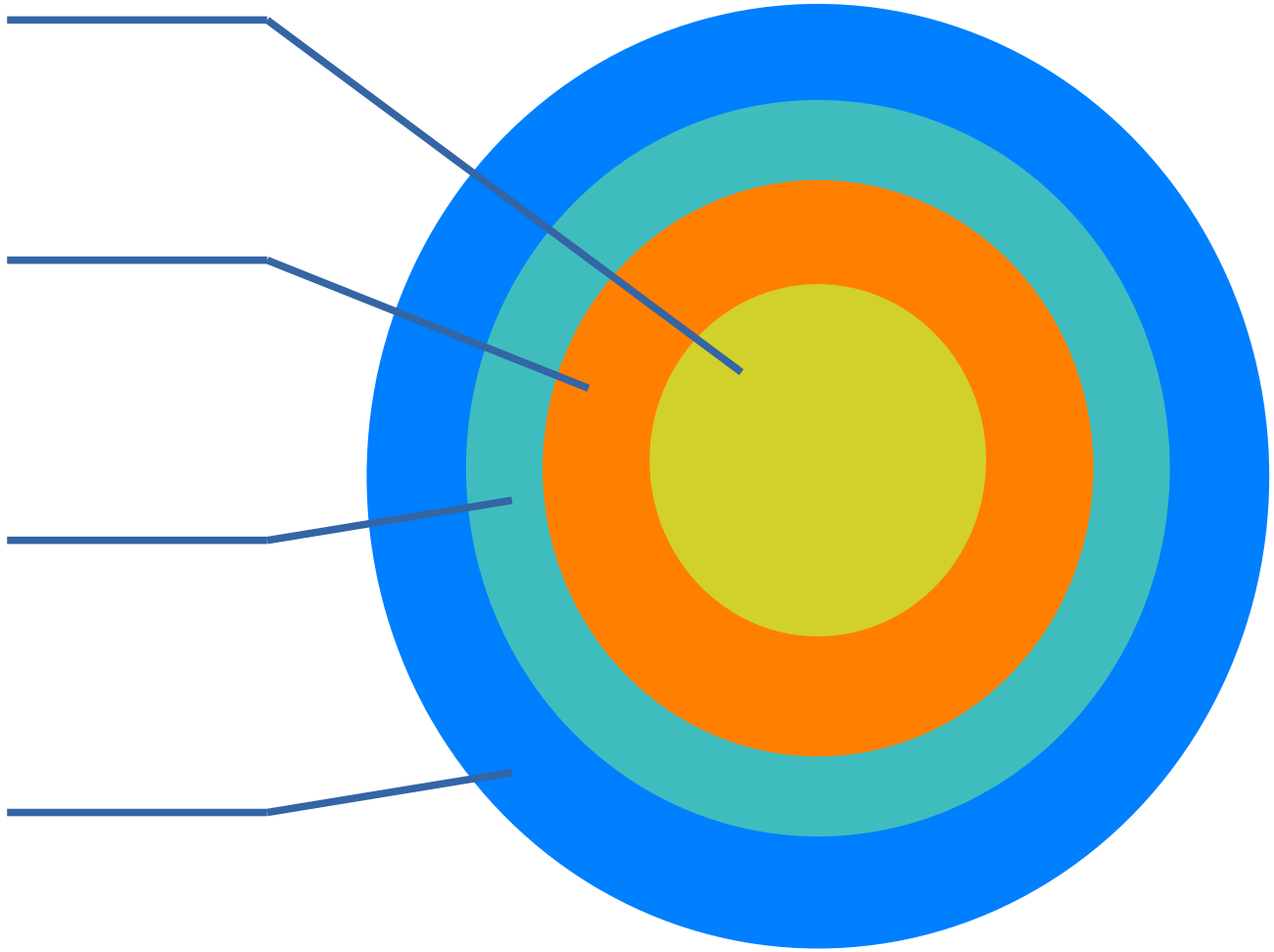
Methodological principle of Digital Health

Multidomain,
continuous data sources

Artificial Intelligence (AI),
Context Awareness

Digital Biomarkers,
Digital Health Twins

Medical Guidance &
Decision Support



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Lab systems

Stationary, non-personalised



Mobile systems

Customisable via (health) apps



Personal systems



Time

Computing shift

Availability	Sporadic	Continuous
Sensor coverage	Lab / clinic	Free-living
Context/behaviour awareness	None	Full

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Medical data integration in Digital Health by examples

Prof. Dr. Oliver Amft

amft@computer.org

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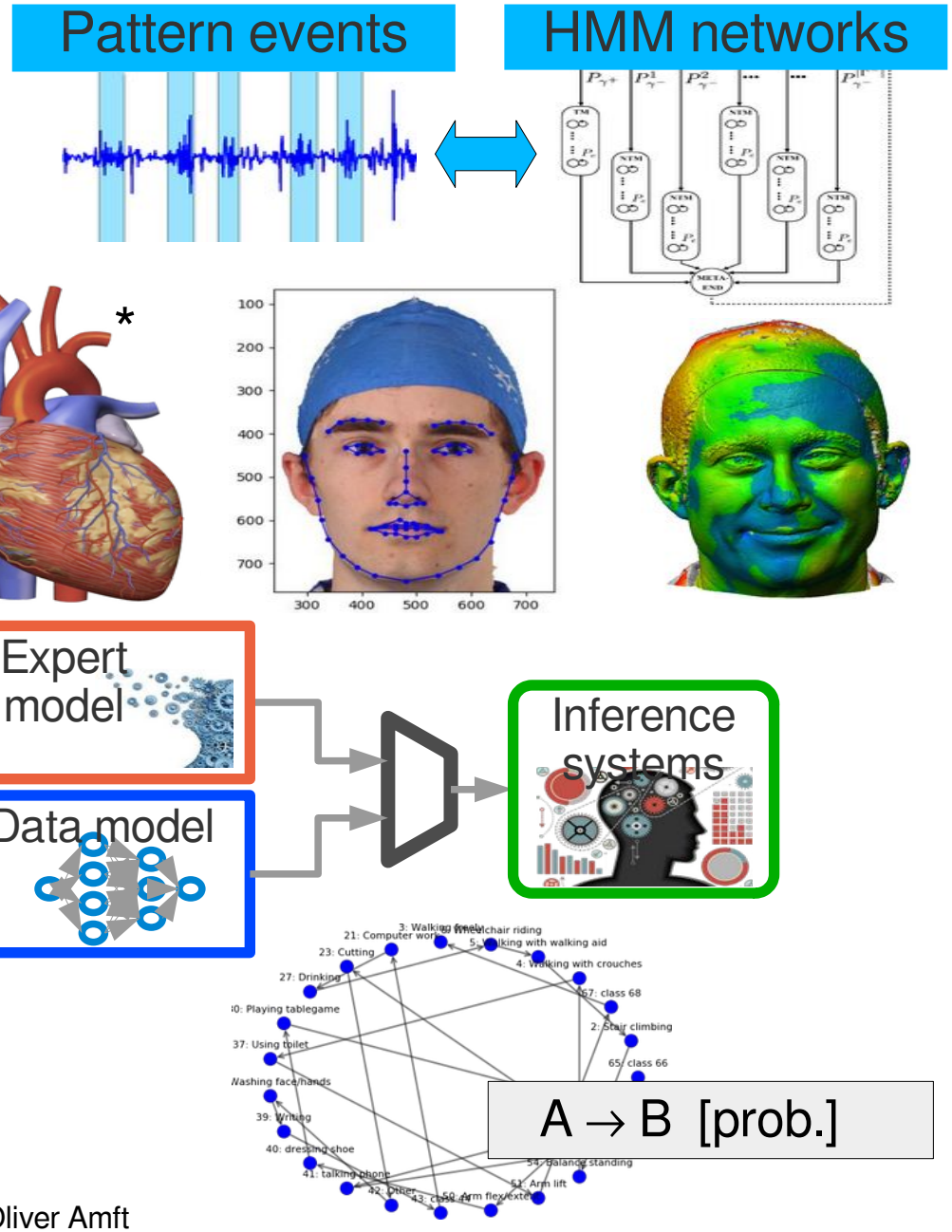
Selected AI tasks in medicine

Pattern spotting.

Image analysis, 3D reconstruction.

Incorporating expert knowledge.

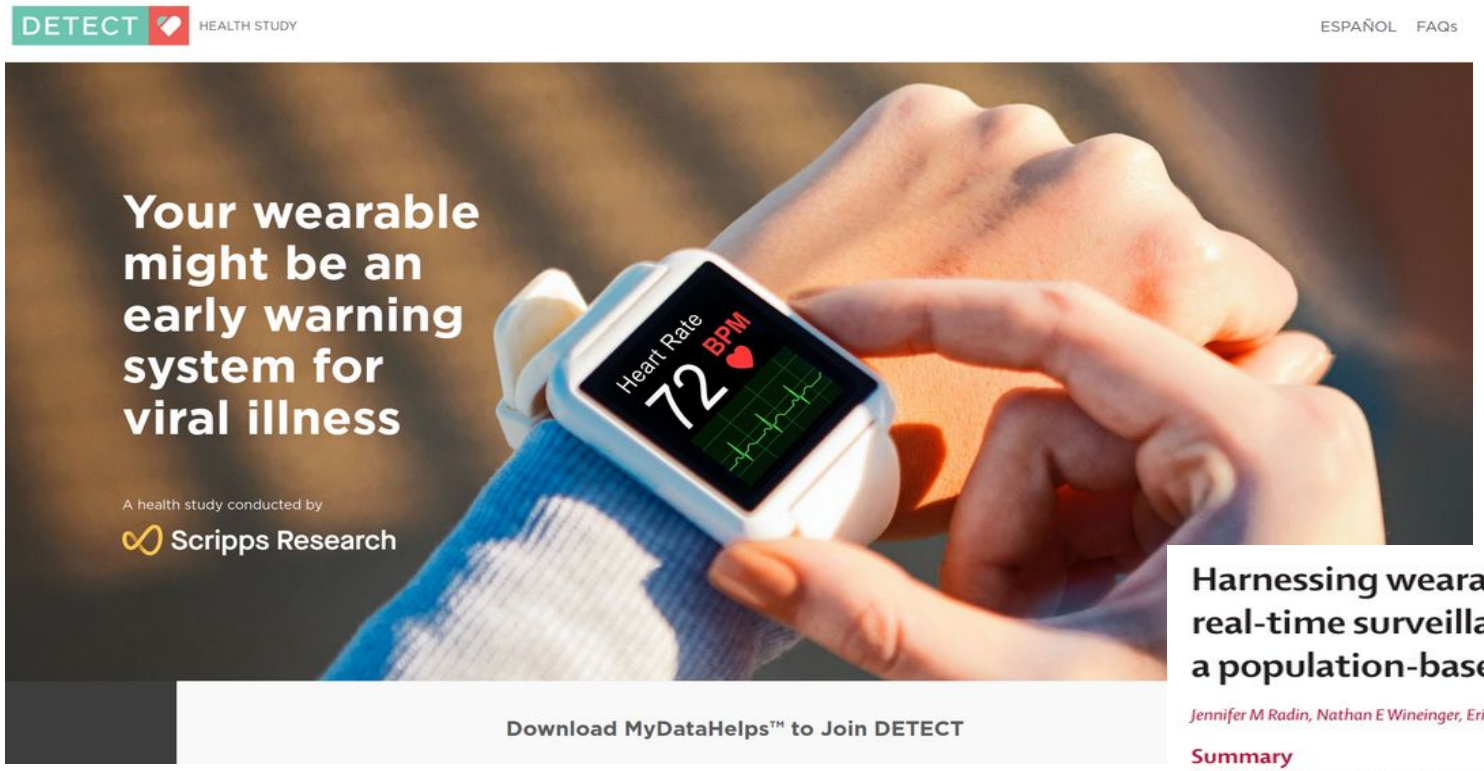
Knowledge mining, rule mining.



* BruceBlaus, CCAS4.0

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Symptom tracking



Harnessing wearable device data to improve state-level real-time surveillance of influenza-like illness in the USA: a population-based study

Jennifer M Radin, Nathan E Wineinger, Eric J Topol, Steven R Steinhubl

Summary

Background Acute infections can cause an individual to have an elevated resting heart rate (RHR) and change their routine daily activities due to the physiological response to the inflammatory insult. Consequently, we aimed to evaluate if population trends of seasonal respiratory infections, such as influenza, could be identified through wearable sensors that collect RHR and sleep data.

Methods We obtained de-identified sensor data from 200 000 individuals who used a Fitbit wearable device from March 1, 2016, to March 1, 2018, in the USA. We included users who wore a Fitbit for at least 60 days and used the same wearable throughout the entire period, and focused on the top five states with the most Fitbit users in the dataset: California, Texas, New York, Illinois, and Pennsylvania. Inclusion criteria included having a self-reported birth year between 1930 and 2004, height greater than 1 m, and weight greater than 20 kg. We excluded daily measurements with missing RHR, missing wear time, and wear time less than 1000 min per day. We compared sensor data with weekly estimates of influenza-like illness (ILI) rates at the state level, as reported by the US Centers for Disease Control and Prevention (CDC), by identifying weeks in which Fitbit users displayed elevated RHRs and increased sleep levels. For each state, we modelled ILI case counts with a negative binomial model that included 3-week lagged CDC ILI rate data (null model) and the proportion of weekly Fitbit users with elevated RHR and increased sleep duration above a specified threshold (full model). We also evaluated weekly change in ILI rate by linear regression using change in proportion of elevated Fitbit data. Pearson correlation was used to compare predicted versus CDC reported ILI rates.

Lancet Digital Health 2020; 2: e85–93
Published Online January 16, 2020 https://doi.org/10.1016/S2589-7500(19)30222-5
See Comment page e54

Translational Institute, Scripps Research, La Jolla, CA, USA
(J M Radin PhD, N E Wineinger PhD, Prof E J Topol MD, S R Steinhubl MD)
Correspondence to: Dr Jennifer M Radin, Translational Institute, Scripps Research, La Jolla, CA 92037, USA
jmradin@scripps.edu

Radin et al., Lancet Digital Health 2(2), ‘20

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Corona-Datenspende

ROBERT KOCH INSTITUT



Professor Lothar H. Wieler

Präsident Robert Koch-Institut

"Digitale Anwendungen können die bisherigen Maßnahmen zur Eindämmung von COVID-19 sinnvoll ergänzen. Wir wünschen uns, dass sich viele Menschen beteiligen. Denn je mehr Menschen ihre Daten für eine Auswertung zur Verfügung stellen, desto genauer werden unsere Erkenntnisse zur Verbreitung des Coronavirus."



Professor Dirk Brockmann

Modellierung komplexer Systeme | HU Berlin

„Ihre Datenspende ermöglicht uns die Ausbreitung des Coronavirus besser zu erfassen und die Dunkelziffer der Infizierten drastisch zu verringern. Diese Informationen sind für Epidemiologen unglaublich wertvoll und helfen, bessere Maßnahmen abzuleiten"

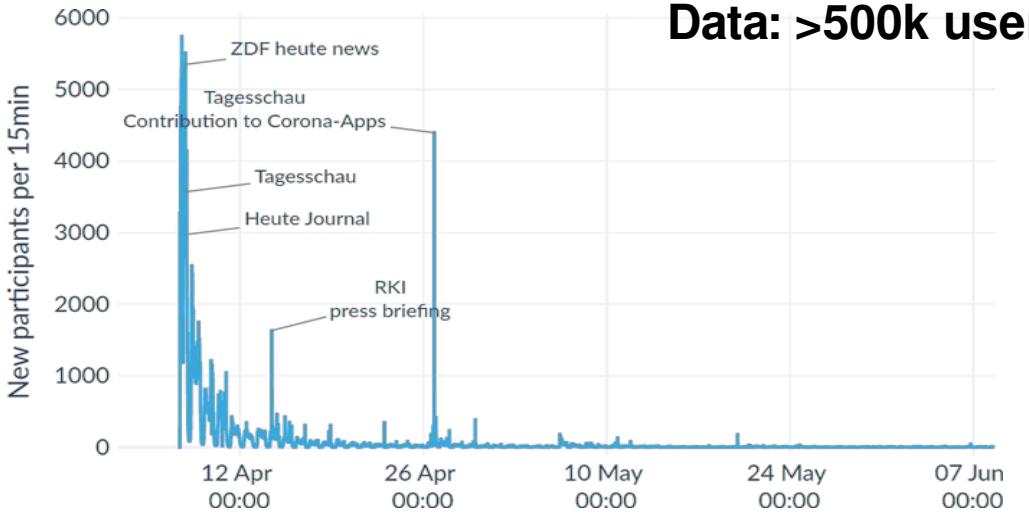


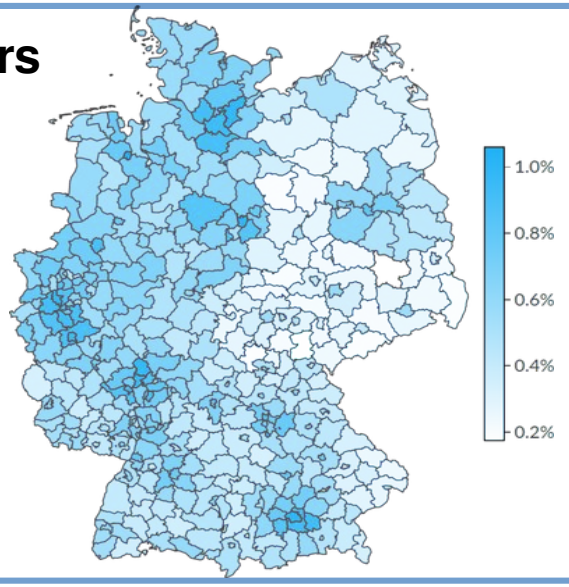
Professor Oliver Amft

Digital Health | FAU Erlangen-Nürnberg

"Zur Vorhersage der Ausbreitung und Eindämmung von COVID-19 werden uns Daten von getragenen Sensoren und Smartphones helfen. Indikatoren für eine Virusinfektion aus Sensordaten sind dabei zuverlässiger als die manuelle Eingabe von Krankheitssymptomen ins Smartphone."

Data: >500k users

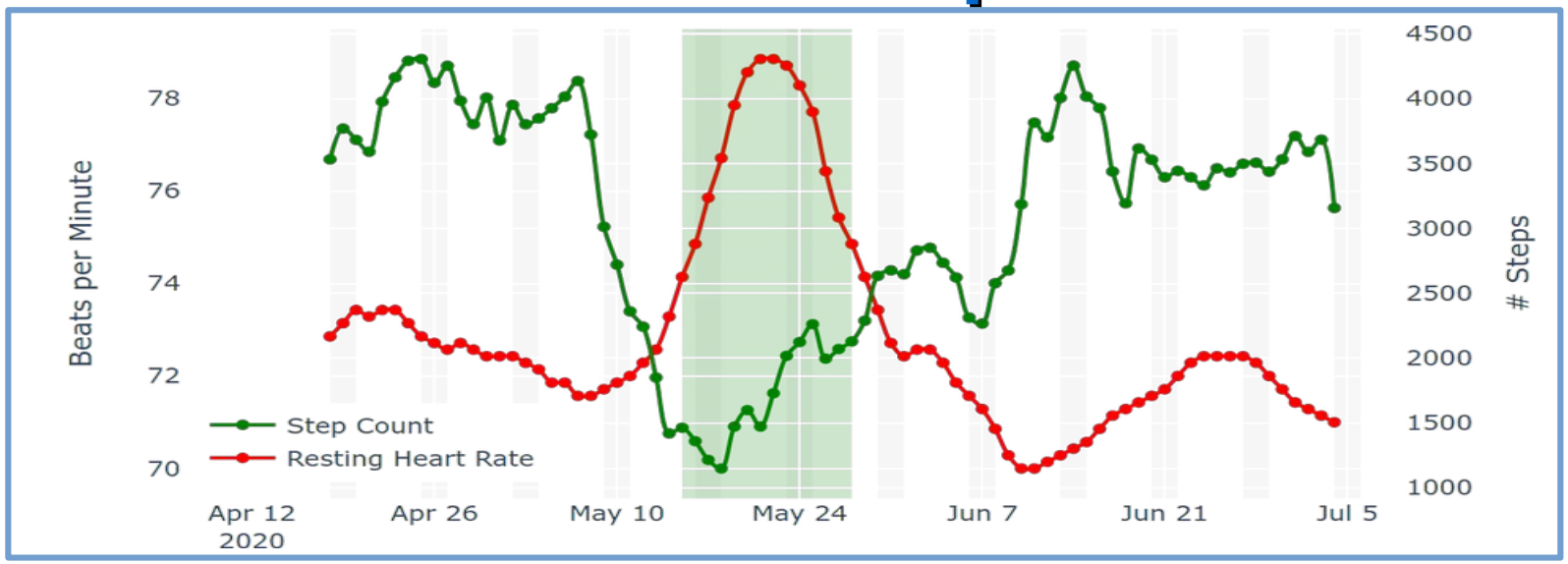


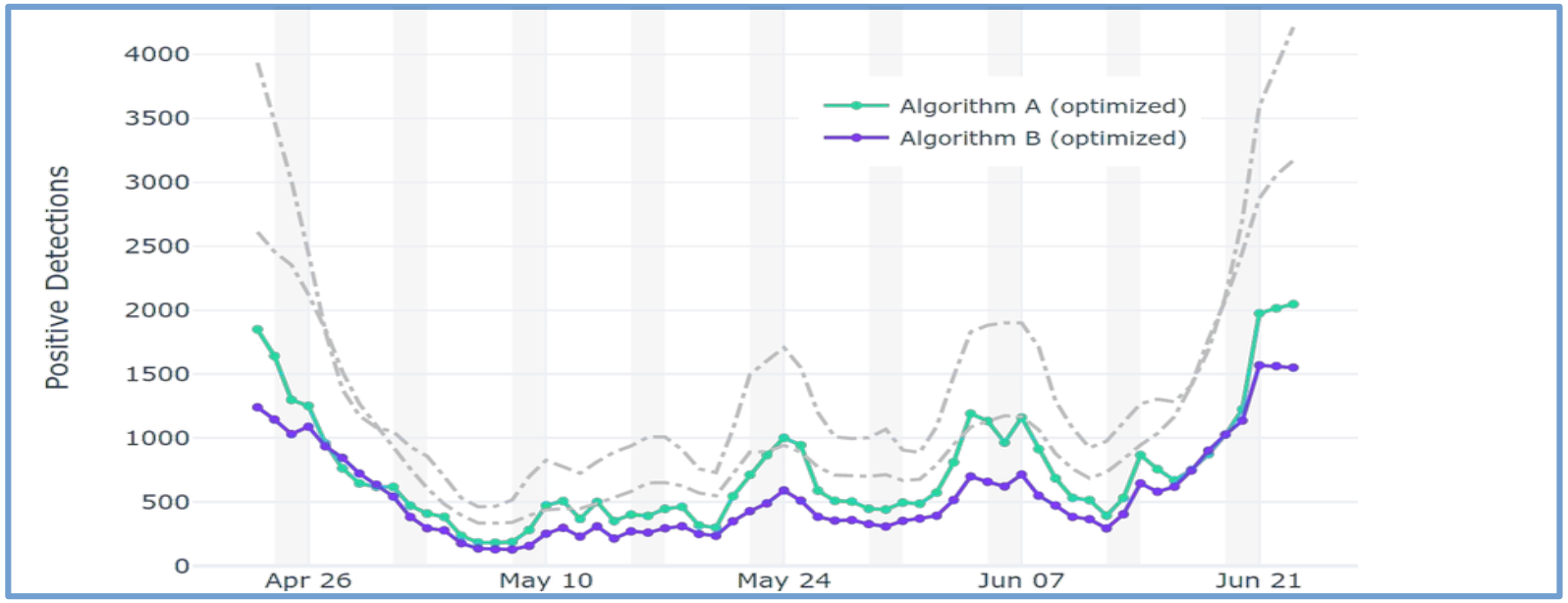


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https://corona-datenspende.de

RHR and step count

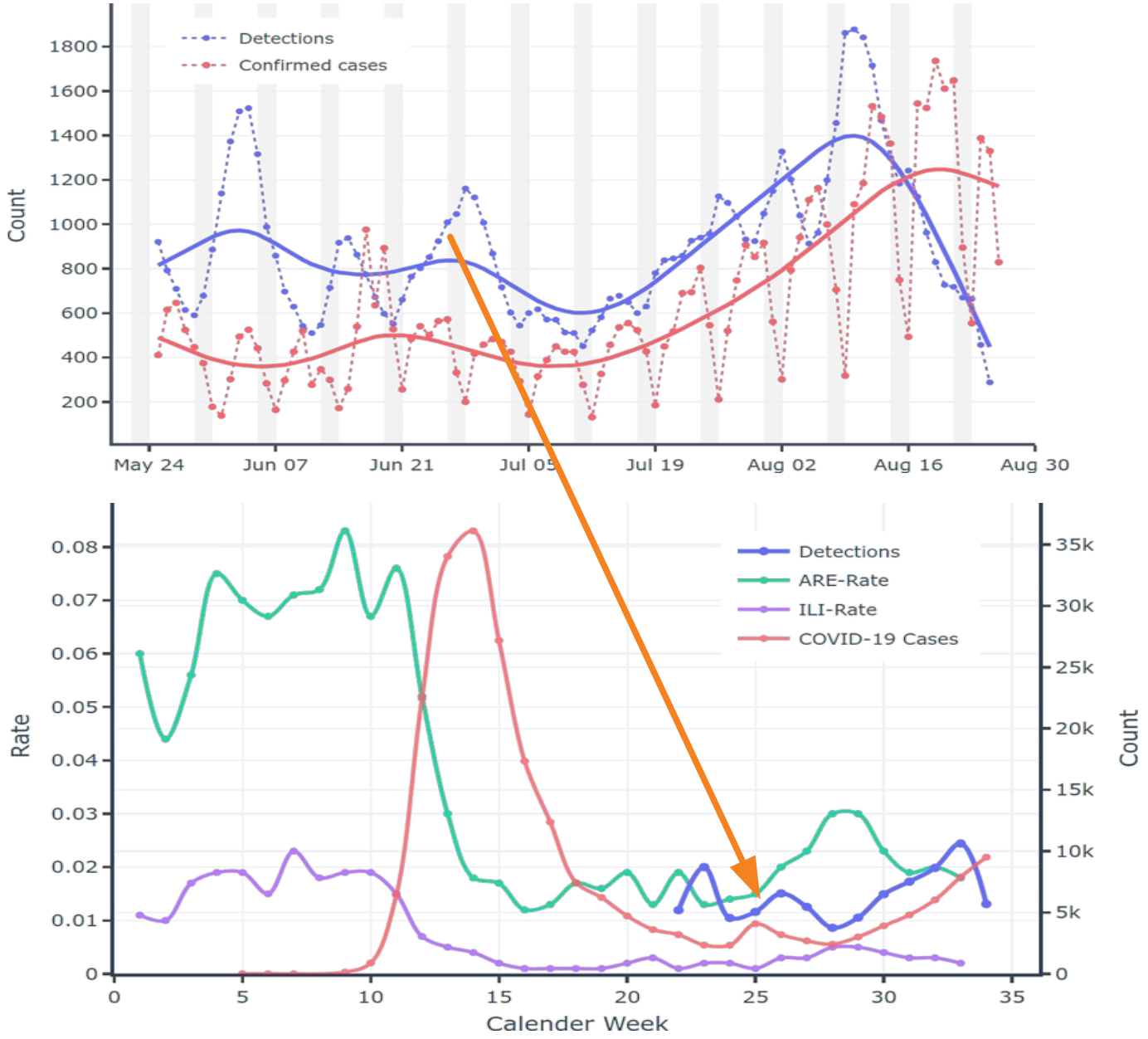




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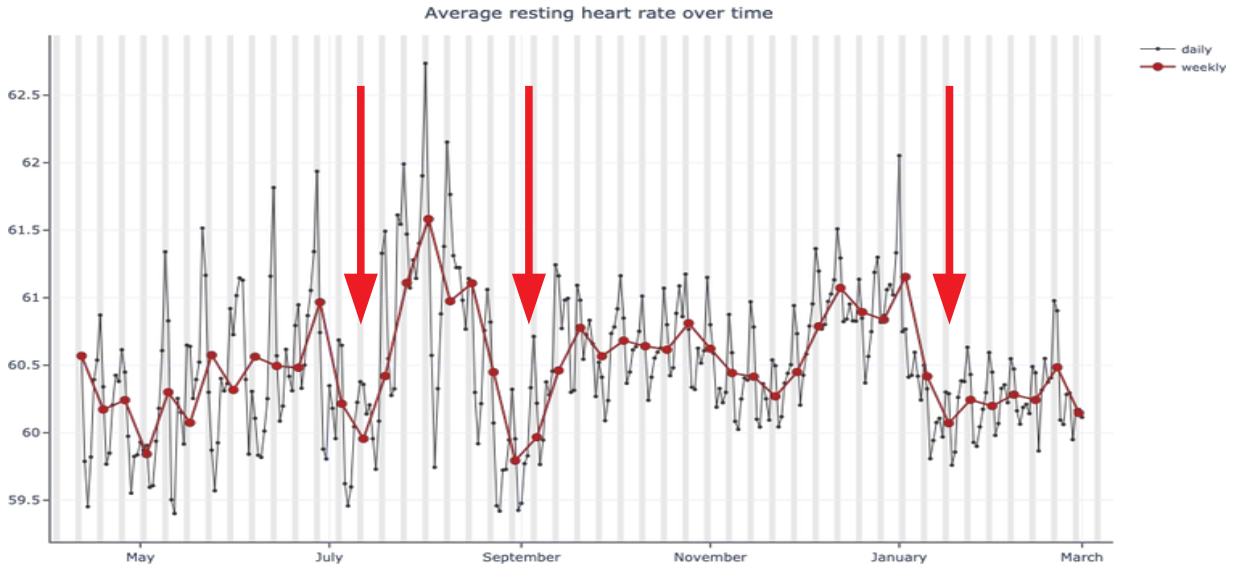
Amft et al., IEEE Perv Comput, 2020

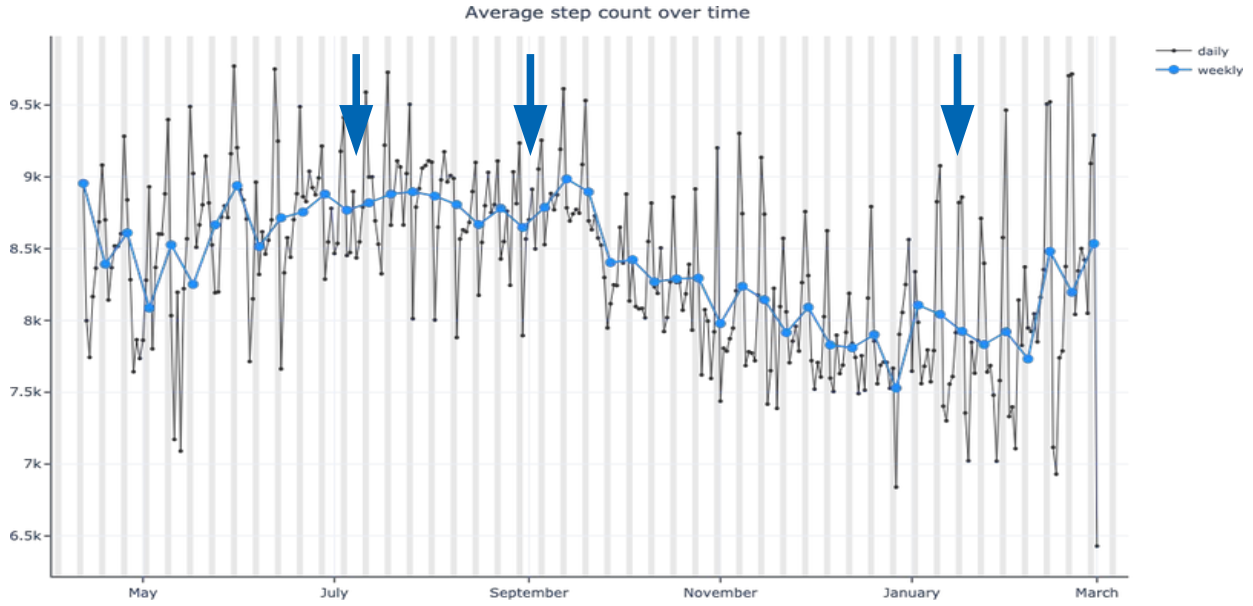
Data trends



https://corona-datenspende.de

The weather issue

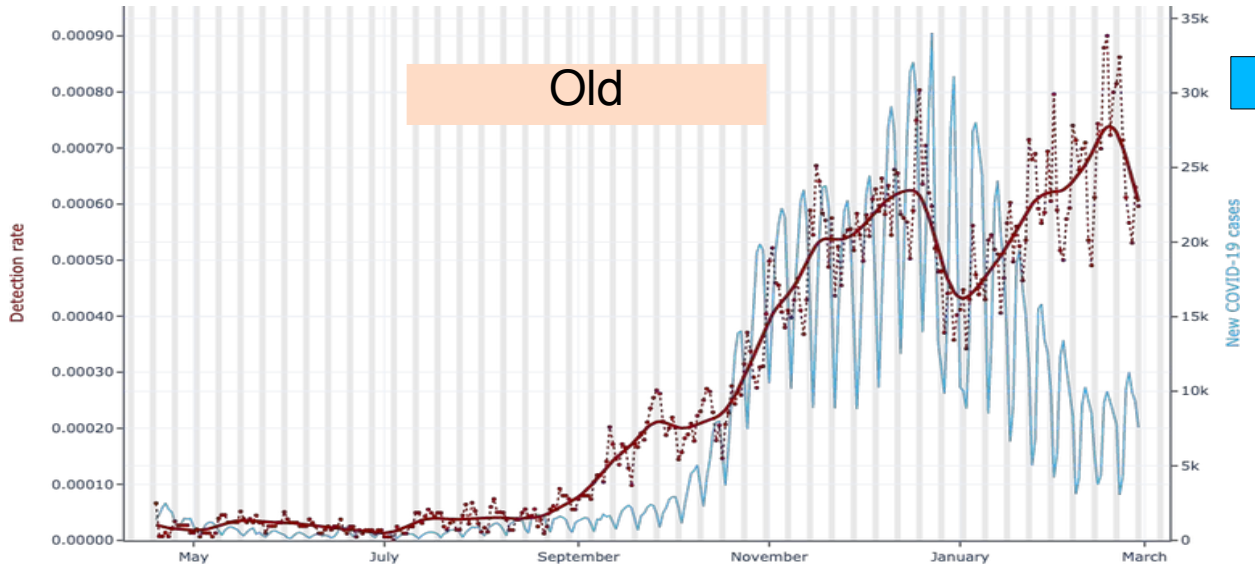


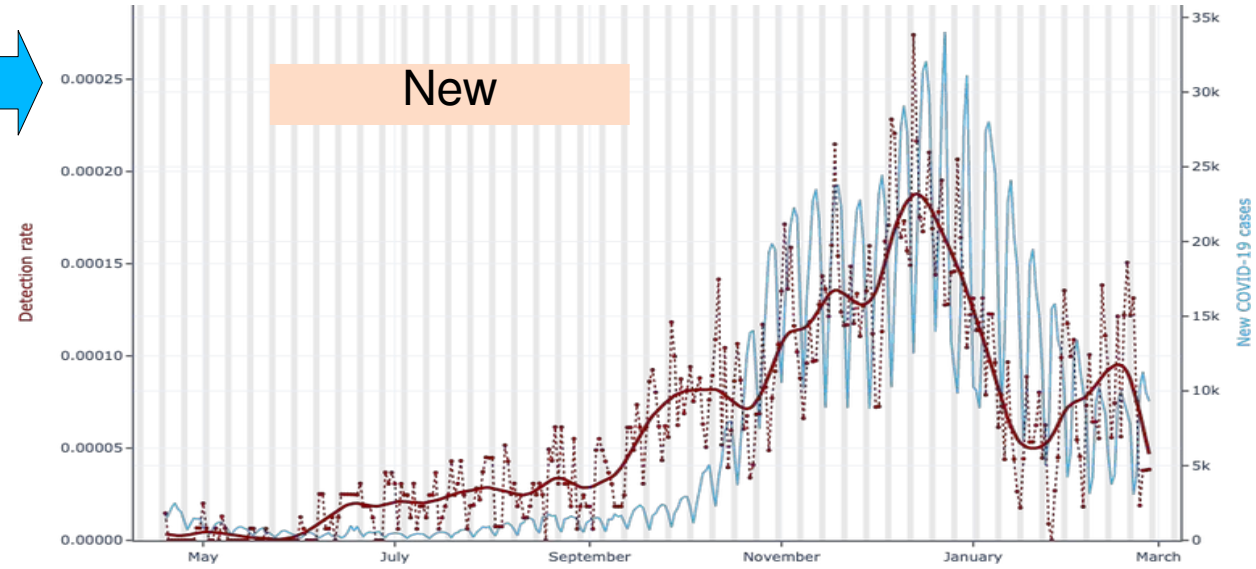


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https://corona-datenspende.de

Normalisation + individual baselines

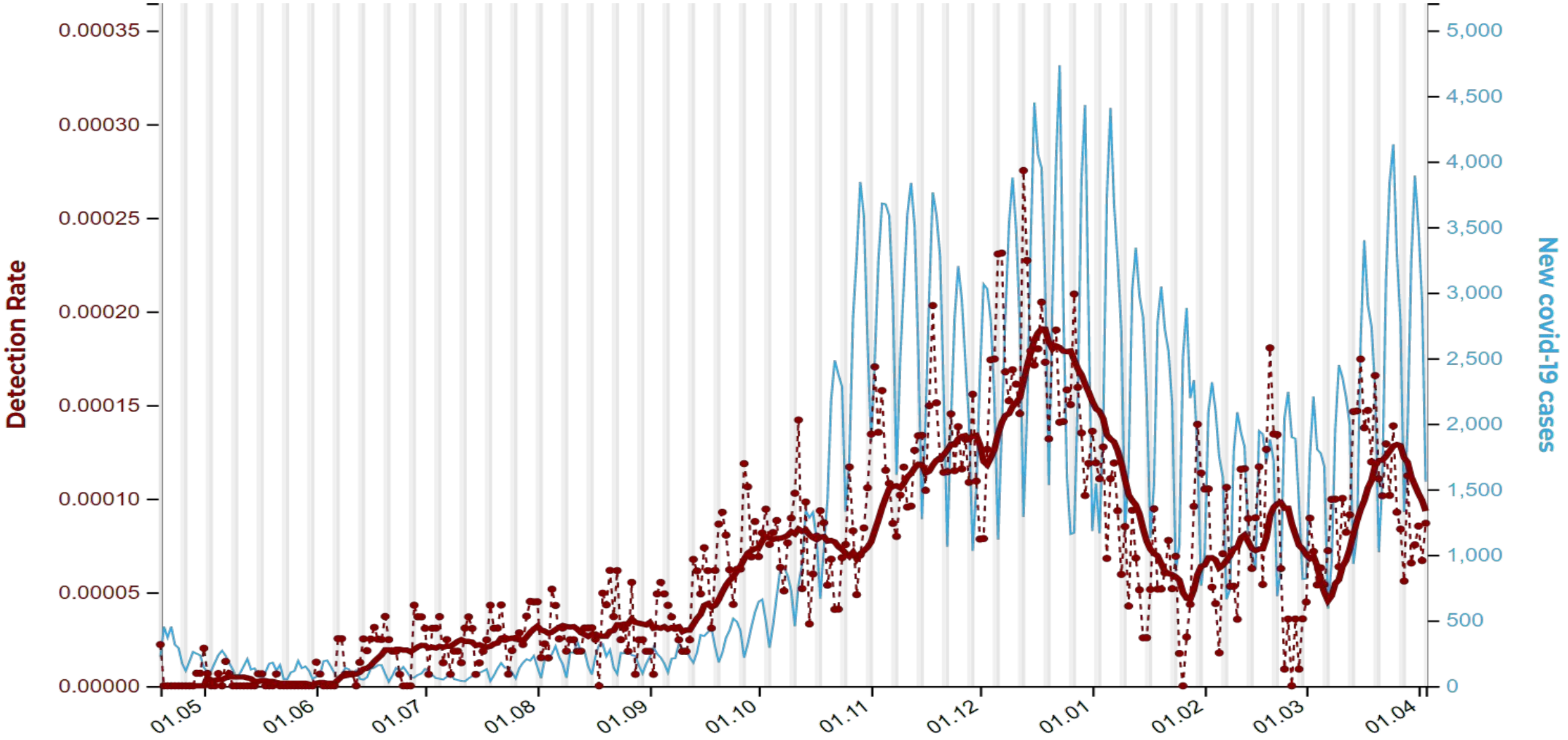




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https://corona-datenspende.de

Fever Curve for Germany



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https://corona-datenspende.de

IEEE pervasive COMPUTING
MOBILE SYSTEMS | UBIQUITOUS COMPUTING | INTERNET OF THINGS

Wearables to Fight COVID-19: From Symptom Tracking to Contact Tracing

Oliver Amft
FAU Erlangen-Nürnberg

Luis Ignacio Lopera González
FAU Erlangen-Nürnberg

Paul Lukowicz
DFKI Kaiserslautern

Sizhen Bian
DFKI Kaiserslautern

Paul Burggraf
Thryve

IEEE Proof

THE NEW CORONAVIRUS pandemic has promoted the new development of mobile and wearable computing in unprecedented ways. We discuss how on-body devices can help to fight the pandemic and may stay as a toolset to effectively deal with infectious diseases in the future.

WHY WEARABLES?

Researchers and health policy managers turned to smartphones and on-body devices mostly for their ubiquity, i.e., to offer health and safety-related information to a large share of the population or gather patient responses. Yet, smartphones and wearables enable fast data and information flow, which is particularly relevant for the rapid infectious character of SARS-CoV-2. We observe that continuous sensor and behavior data of smartphones and on-body devices are important as virus testing is associated with effort, cost, and provides only one-time information, and global immunization is still far away.

We focus here on already existing and newly created wearable devices and smartphone apps for everyday use, but we exclude clinical and laboratory measurement systems, e.g., for heart and respiratory assessment. For researchers and public health authorities, symptom screening and tracking based on continuous sensor data from wearables and smartphones can

Digital Object Identifier 10.1109/MPRV.2020.3021321

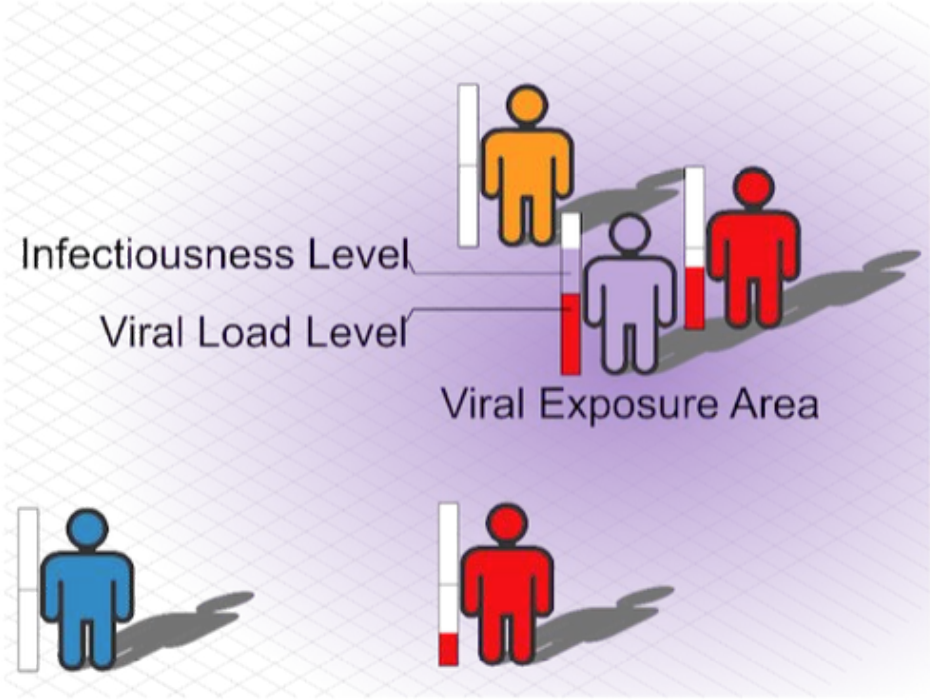
Amft et al., IEEE Perv Comput 19(4), ‘20

Digital Contact Tracing



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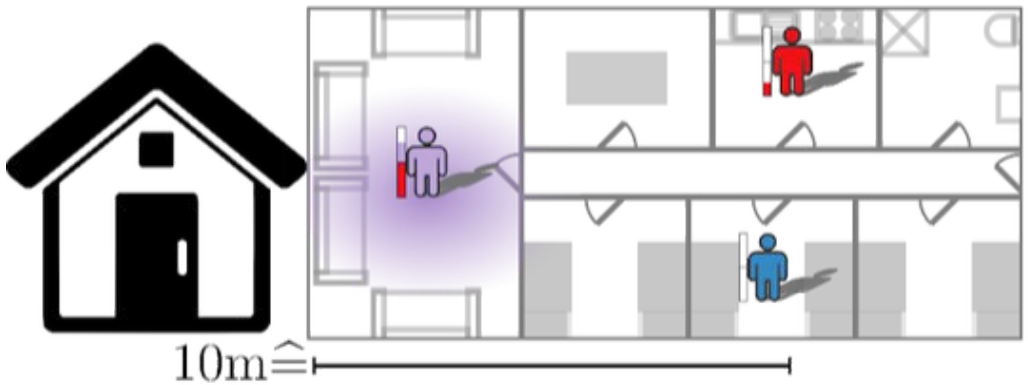


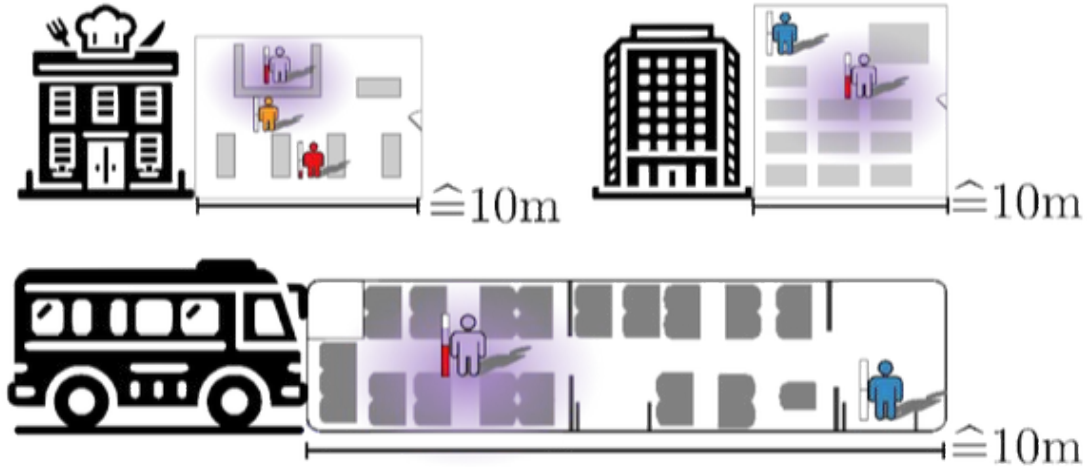


Legends

a)
- Daytime
- Daily routines
- Locations
- Activities

b & c)
- Susceptible
- Exposed
- Infected
- Immune





Lopera et al., 2026

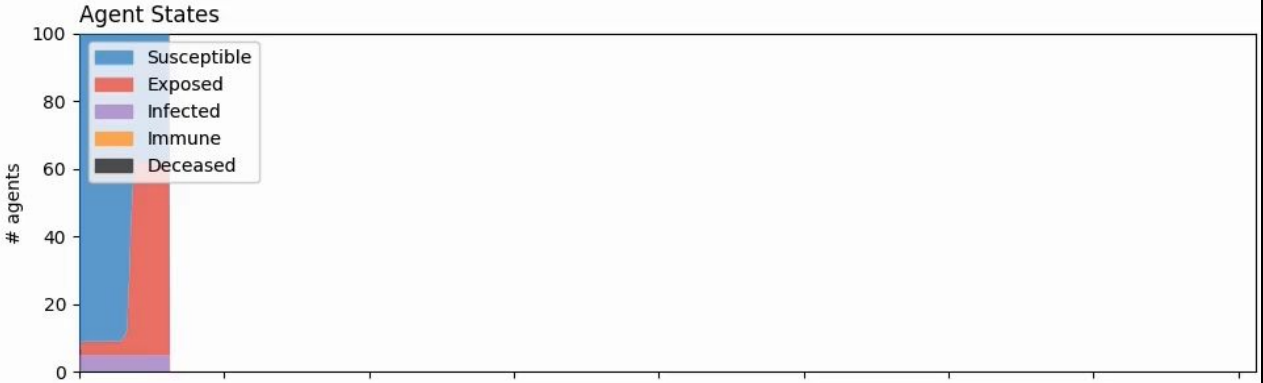
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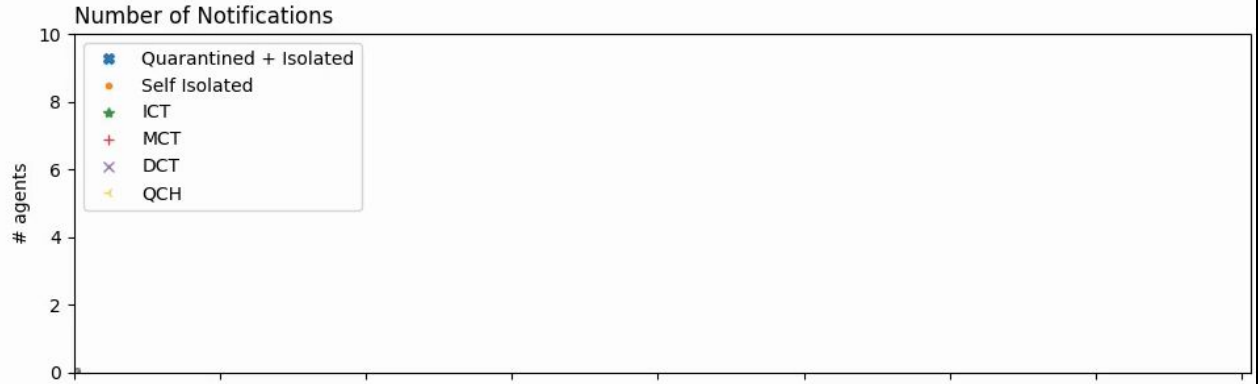
Elapsed time [Days - HH:MM]

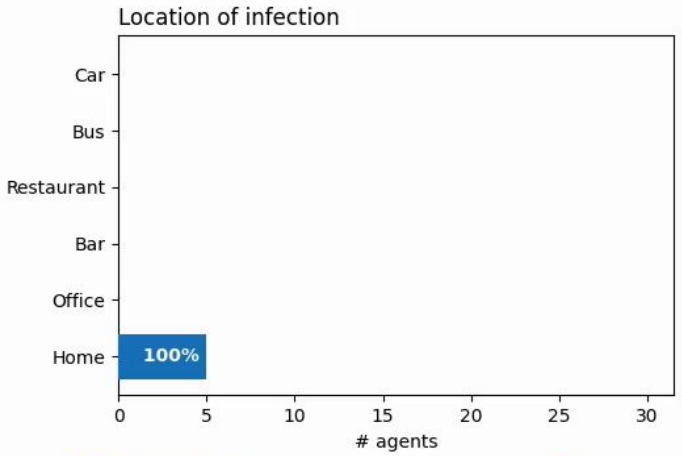
Run Parameters

Contact tracing strategy: ICT, MCT, DCT
DCT compliance rate: 100%
Quarantine contact's household (QCH): False
Request negative test to exit quarantine (RNT): True
Closure of bars and restaurants (CBR): False







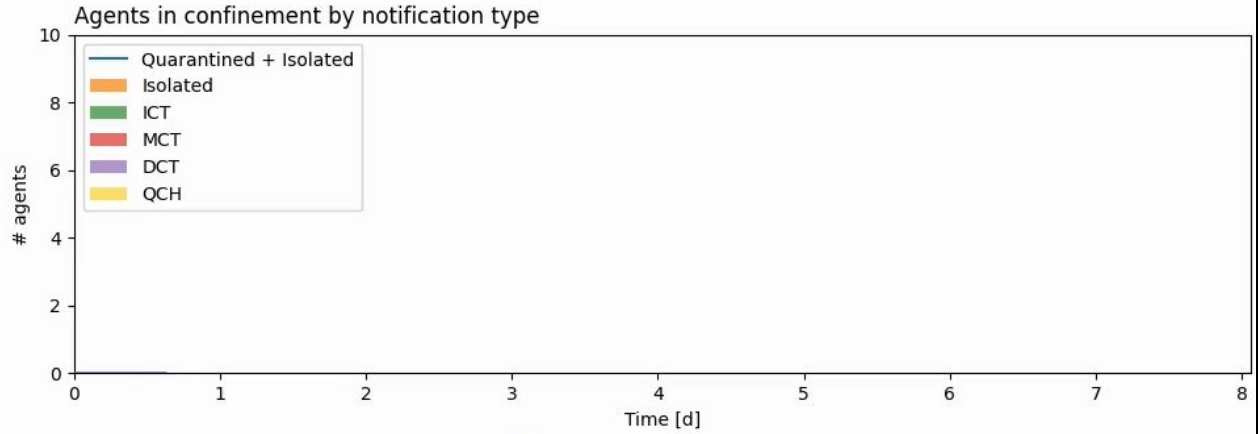


Pandemic characteristics

Pathogen:
Clearance period: 7.6 d
Incubation period: 7.2 d
Generation interval: 0.0 d
Serial interval: 0.0 d

Population:
7-day hosp. incidence: 0.0 d
7-day incidence: 0.7 d
Affected population: 5

Agent:
Avg. Days in quarantine: 0.0 d
Avg. Days in isolation: 0.0 d
Avg. confinements: 0.0 d
False positive rate: 0.0%
False negative rate: 100.0%
Currently confined: 0



Friedrich-Alexander-Universität Medizinische Fakultät

Digital Health

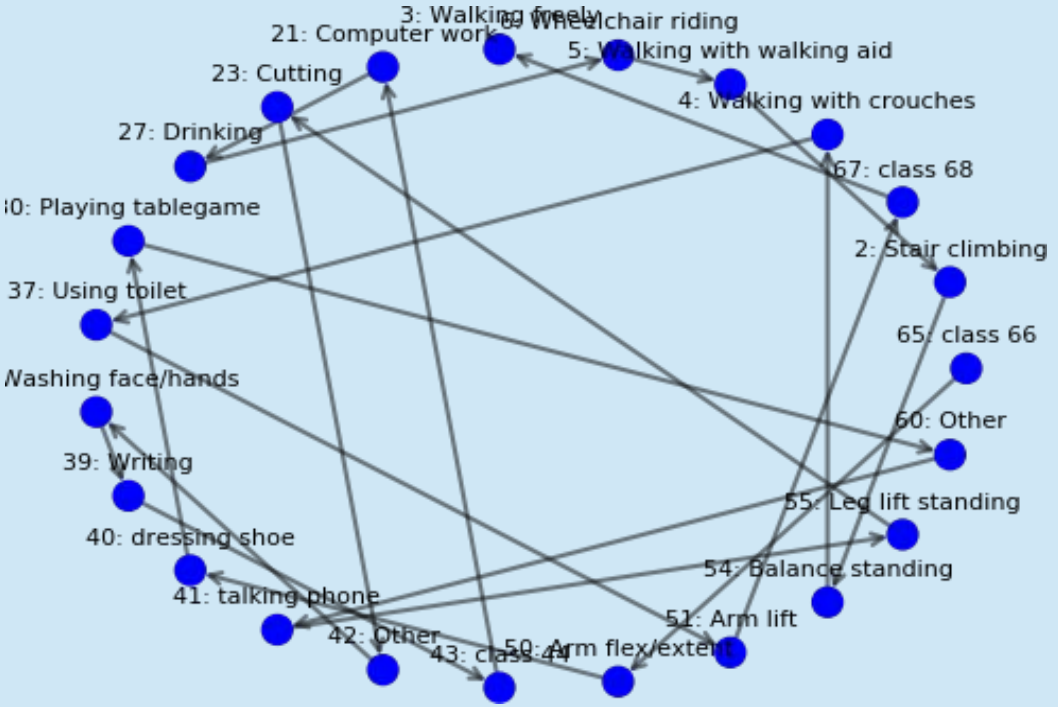
Hahn Schickard

universität freiburg

Intelligent Embedded Systems Lab

ROBERT KOCH INSTITUT

Mining from data



Digital Health



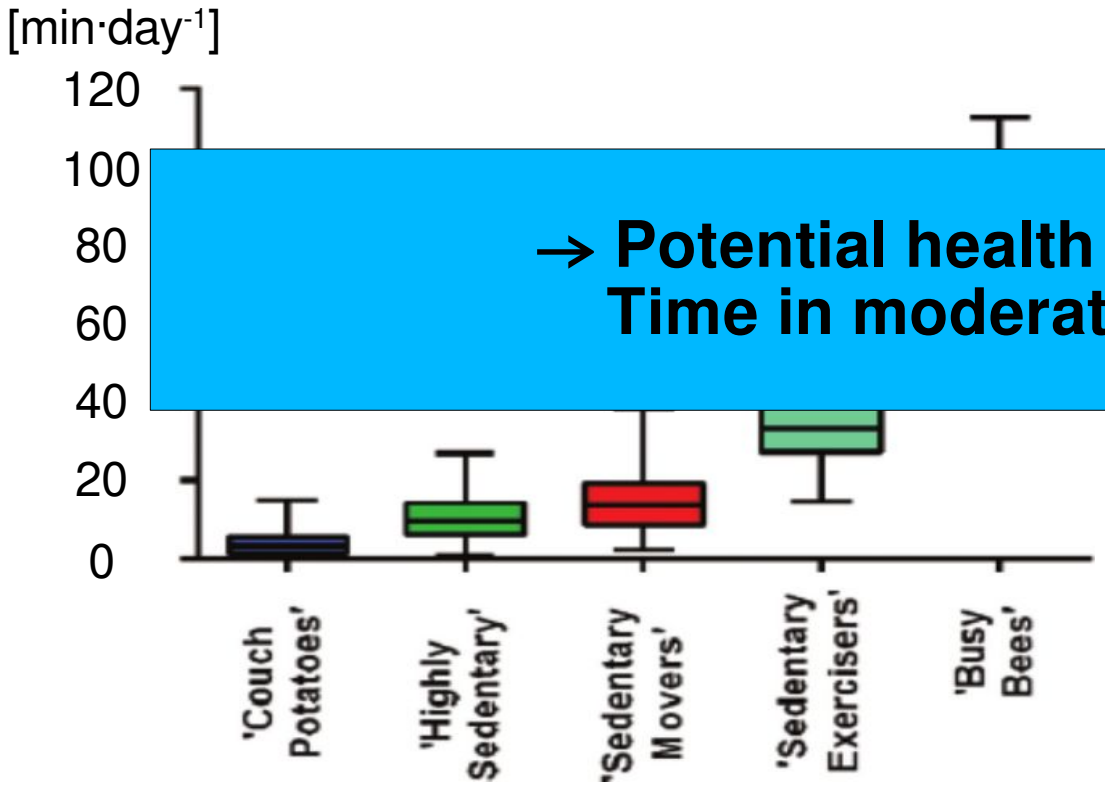
Data: Free-living patients with COPD, ~5-6 days, N=1001

Mesquita et al., Chon Res Dis 14(3), ‘17

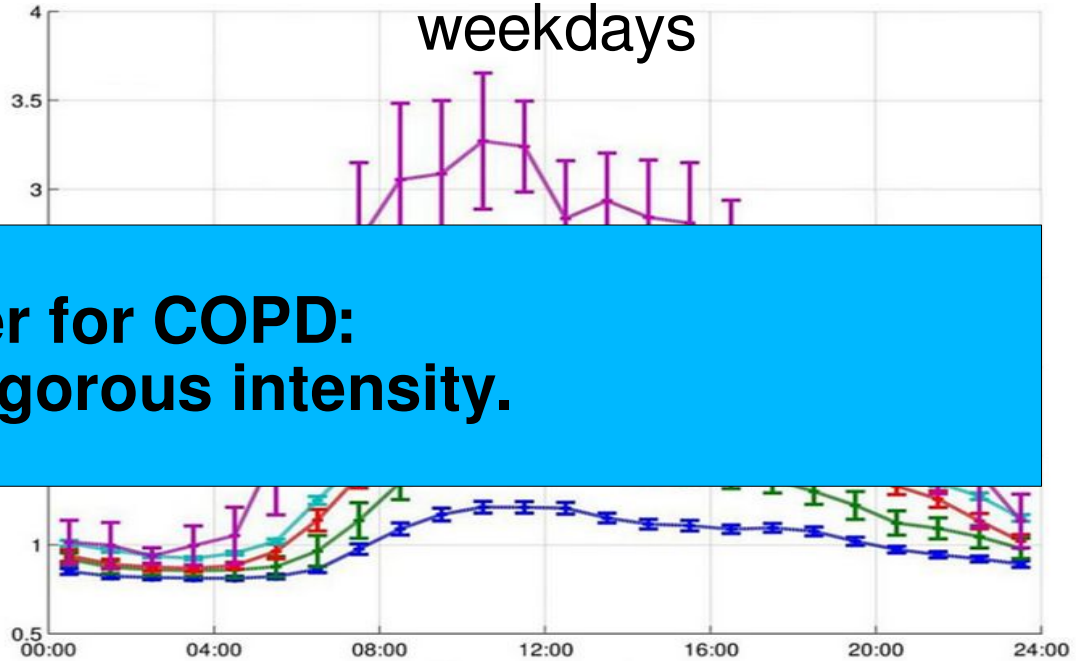


COPD analysis

Time in moderate to vigorous intensity activity



Daily activity patterns during weekdays



→ **Potential health marker for COPD: Time in moderate to vigorous intensity.**

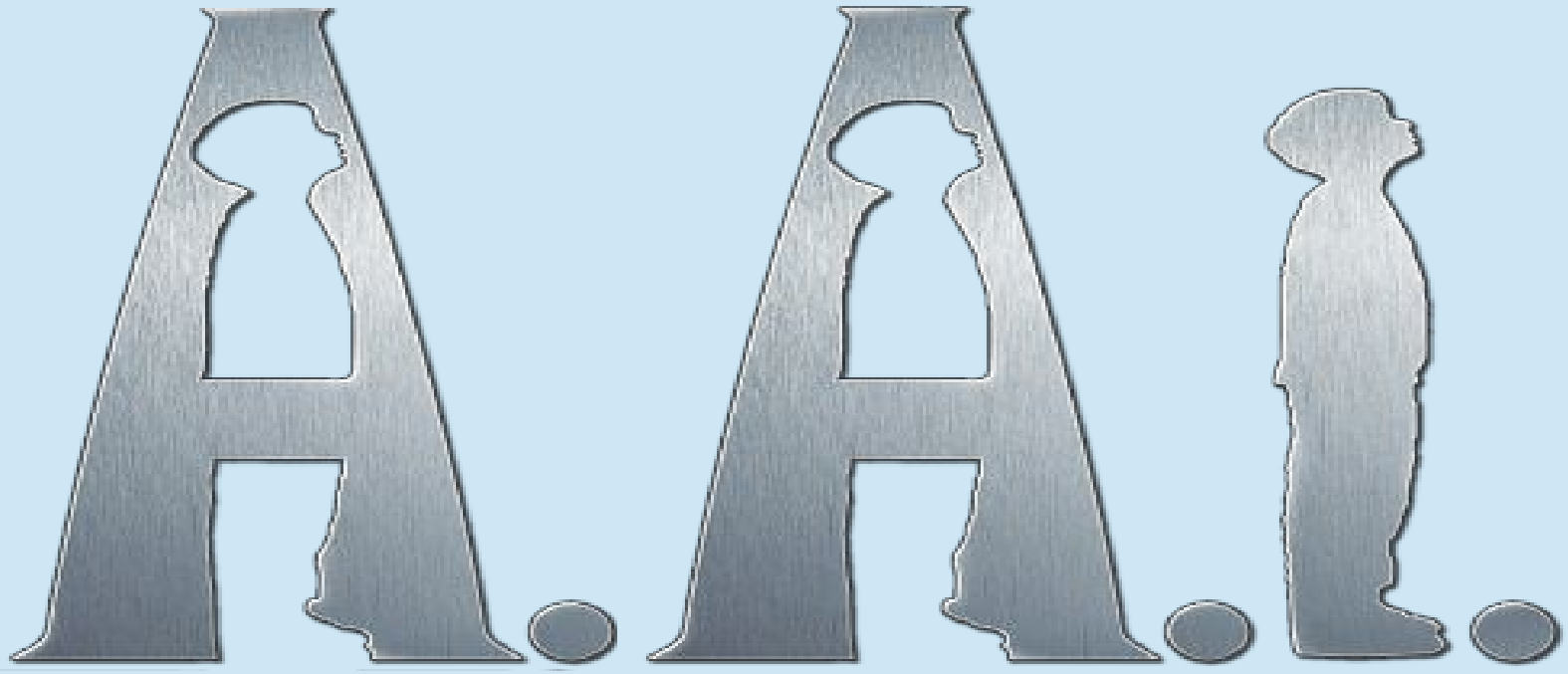
Data: Free-living patients with COPD, ~5-6 days, N=1001

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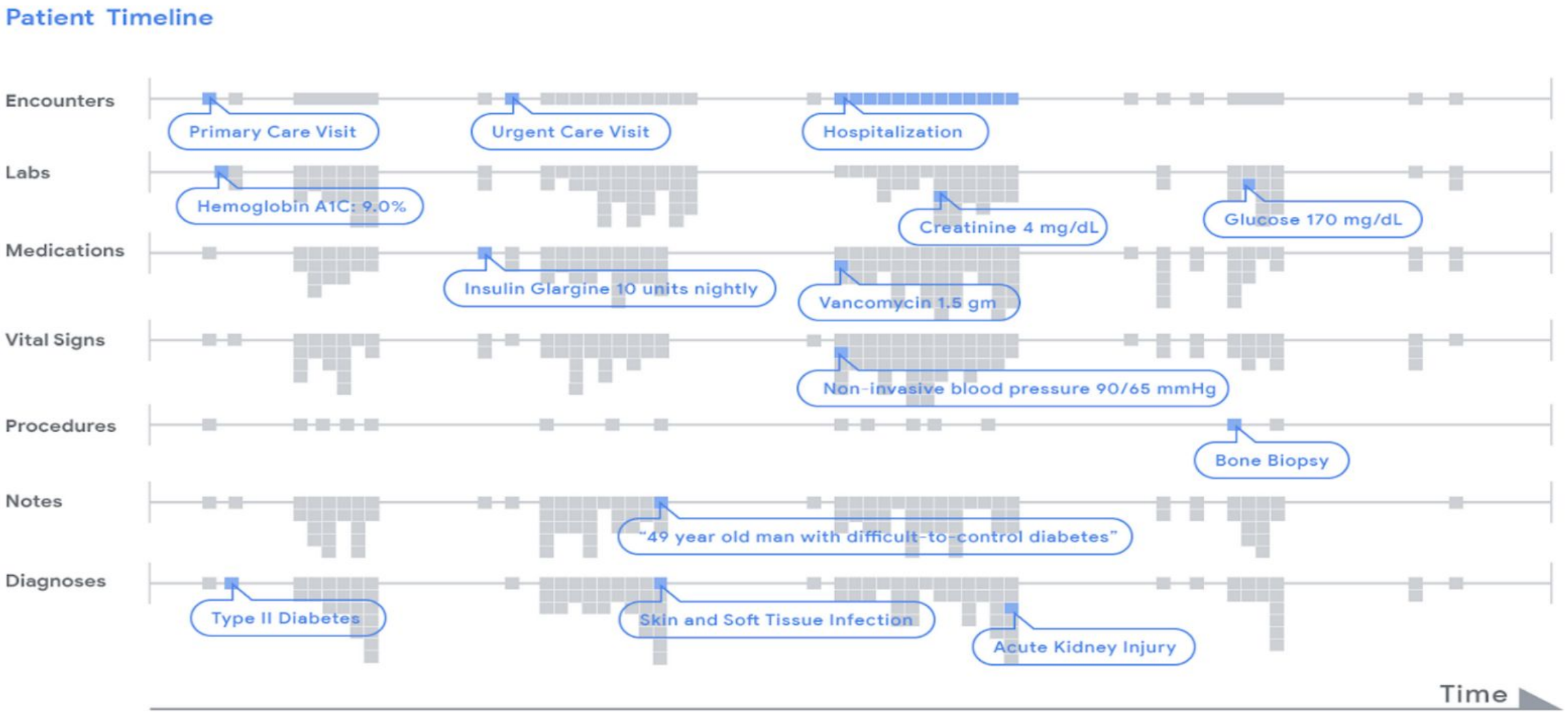
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Mesquita et al., Chon Res Dis, 2017

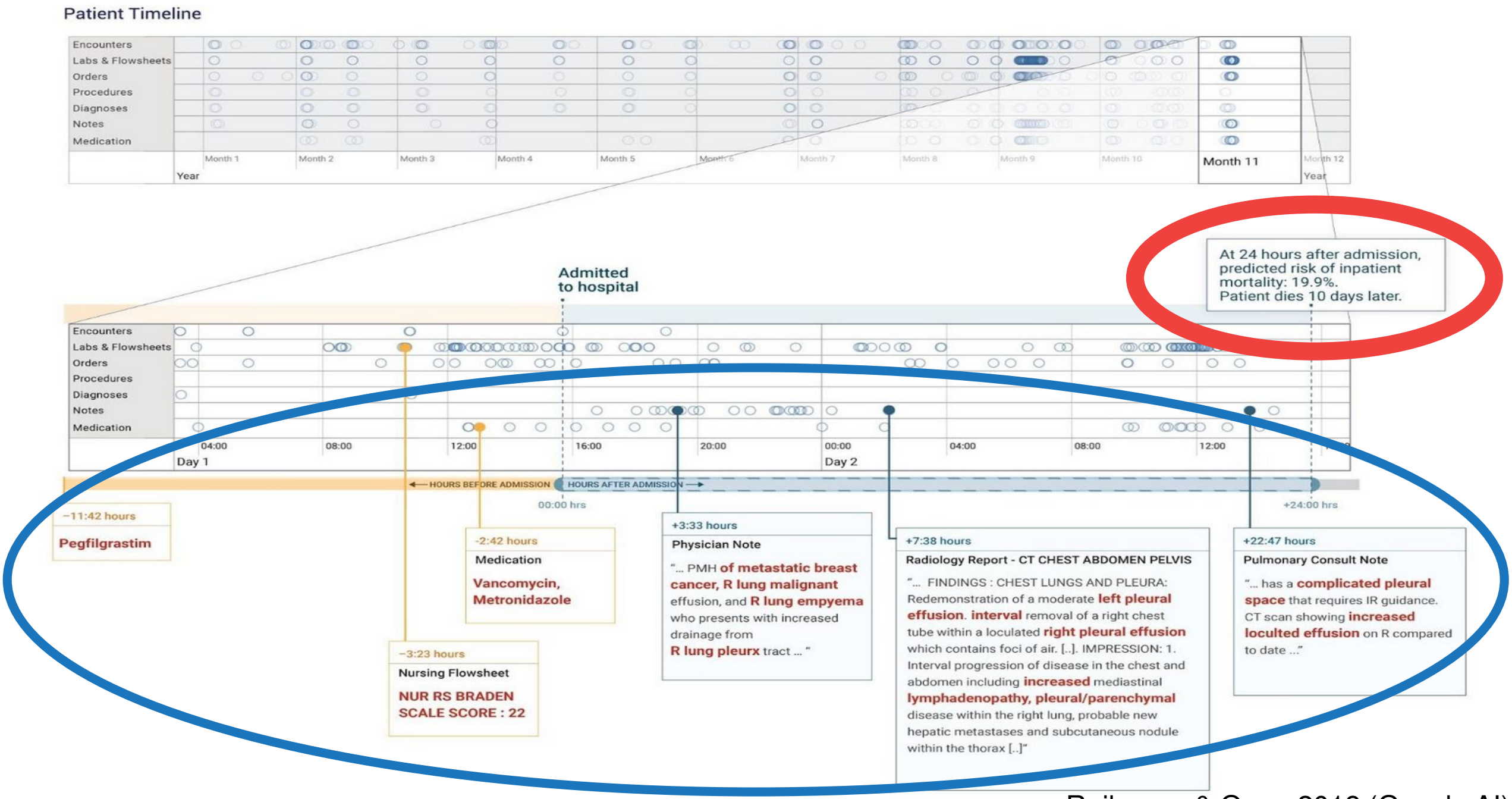
Augmented
Human-Artificial Intelligence



Electronic Health Records

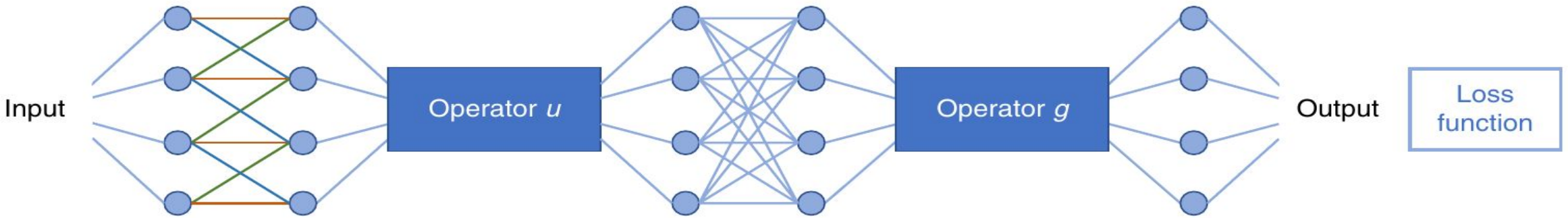


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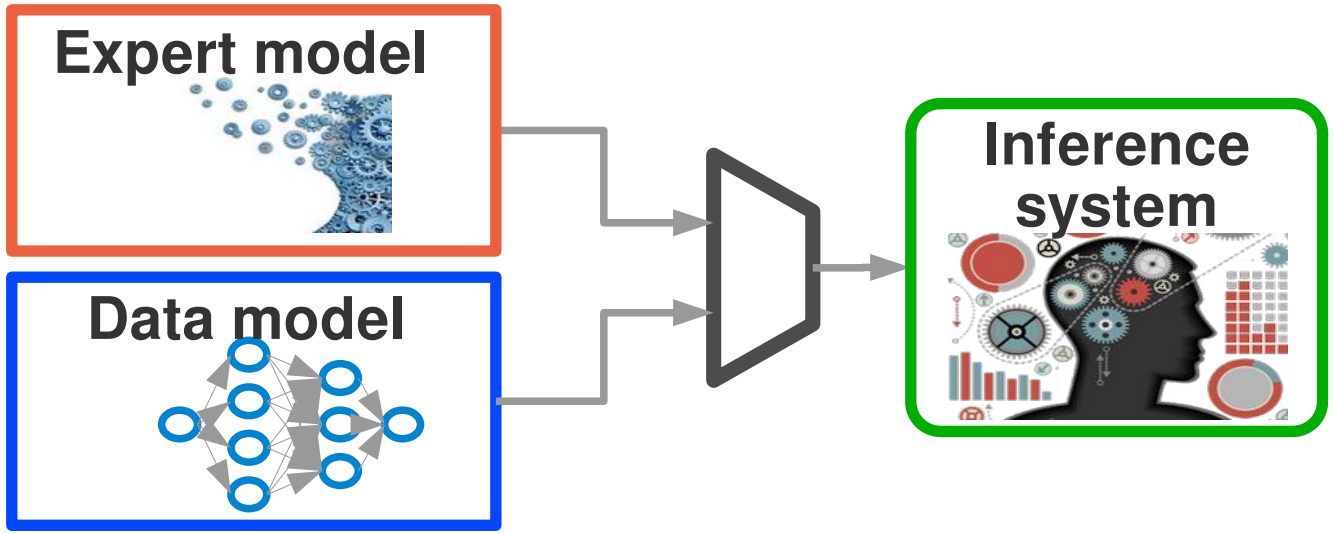


Rajkomar & Oren, 2018 (Google AI)

Incorporating experts' knowledge



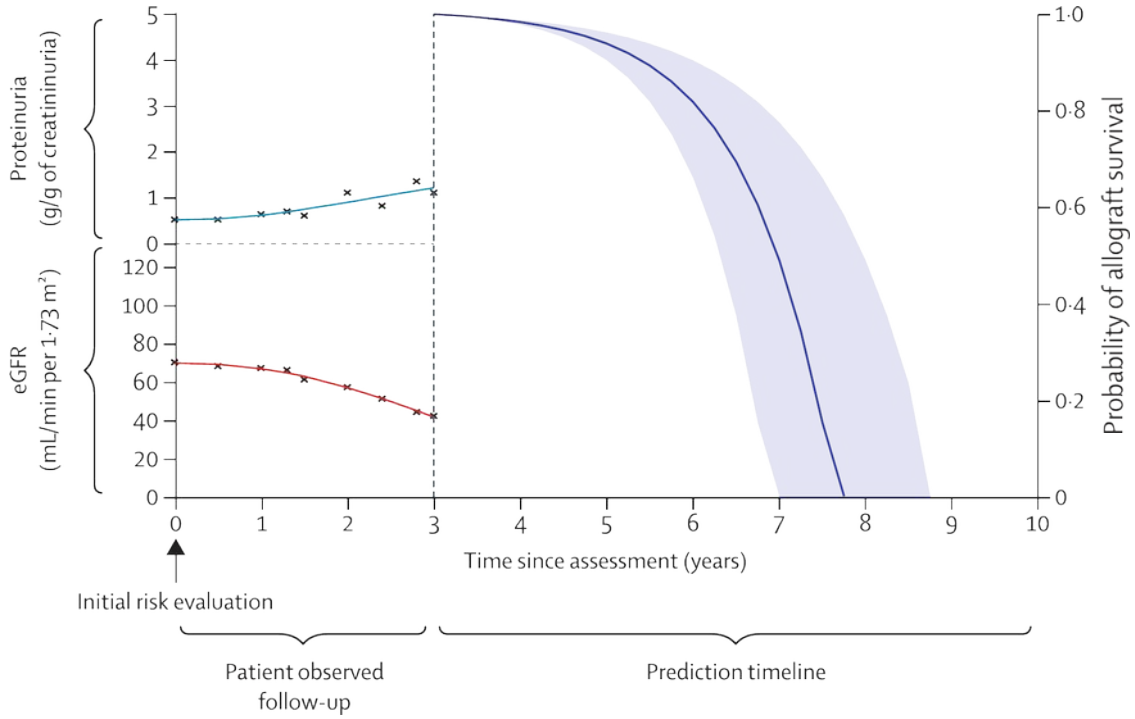
Known operators (here u, g) are embedded into a network to reduce learning parameters. (Maier et al., 2019)



Dynamic fusion of data and expert models. (Wahl & Amft, 2018)

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Does AI add value?



Endpoint		*Subject Groups*	
		Junior MDs	*Senior MDs*
Rejection	*MD*	*0.5714*	***0.7134***
	MD+AI	***0.7553***	*0.4870*
Graft Loss	*MD*	*0.8067*	***0.8108***
	MD+AI	***0.8276***	*0.7320*

Insights:

→ Estimation performance does not always improve!

→ Interaction of experts and AI recommended, but how?

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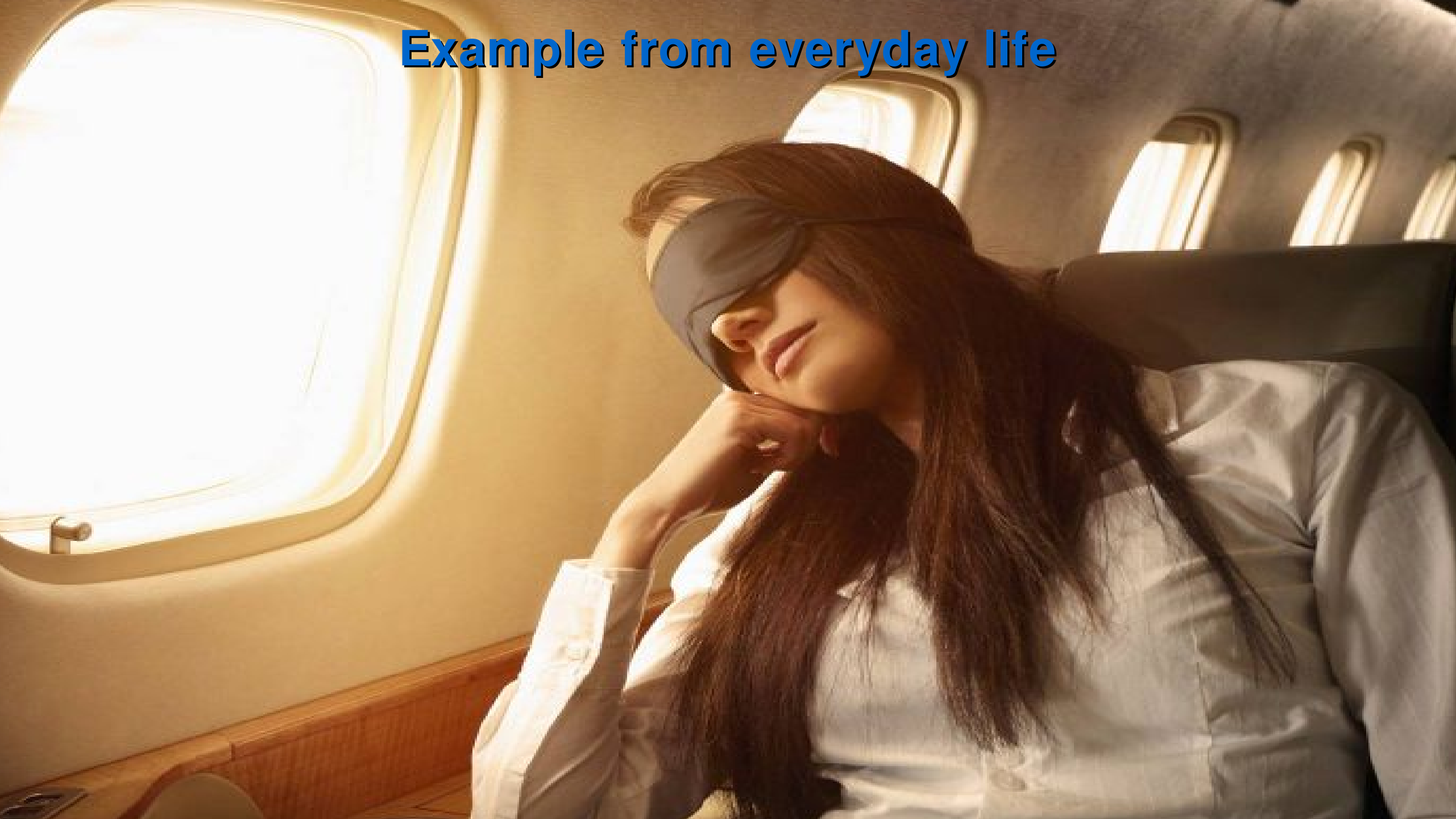
Roller et al., medRxiv, 2022

Context Awareness: Definition

Prof. Dr. Oliver Amft

amft@computer.org

Example from everyday life



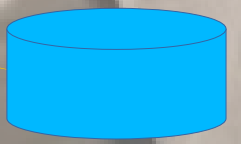
Circadian rhythm



Context awareness

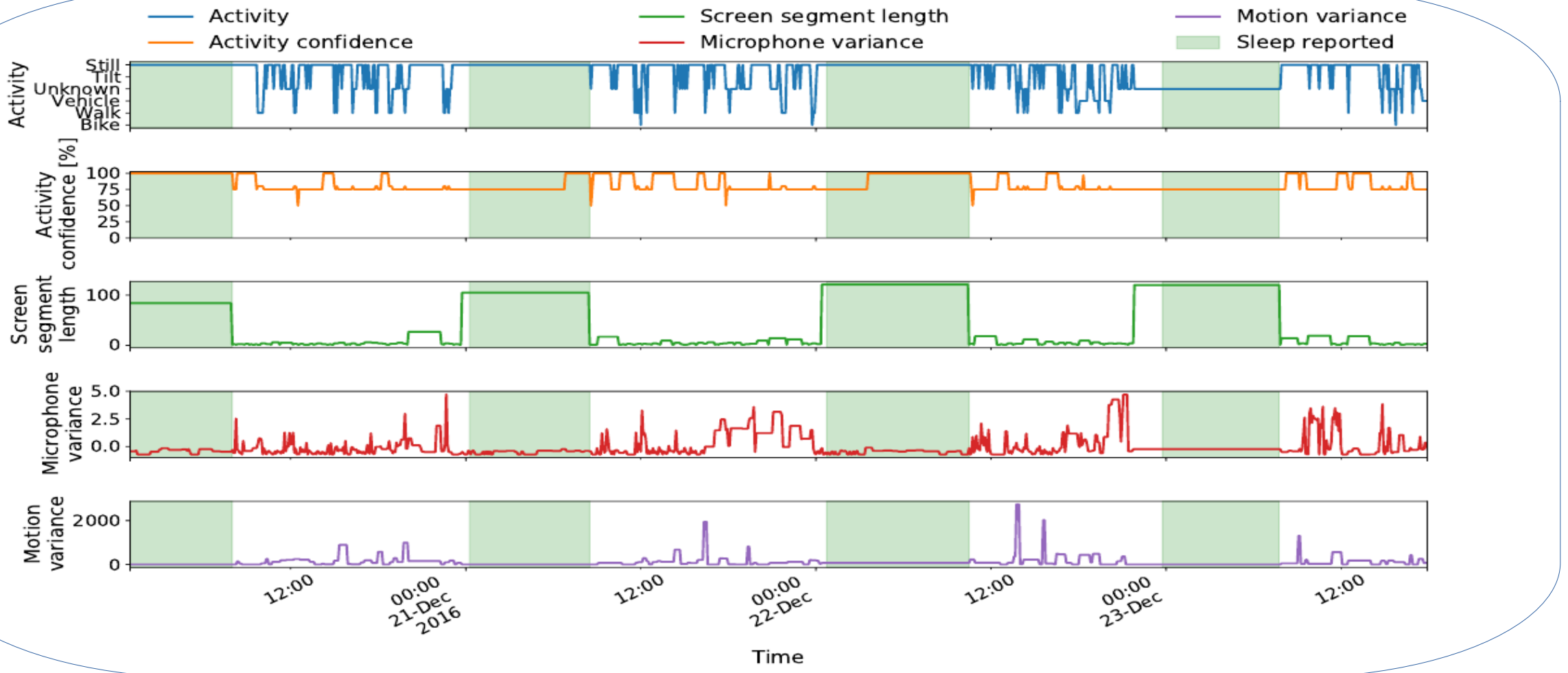


query sensors & profile data



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Context sampling on smartphone





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Context definitions

“Context is the set of environmental states and settings that either determines an application's behavior or in which an application event occurs and is interesting to the user” (Chen & Kotz, 2000)

“any information that can be used to characterise the situation of entities.” (Dey et al., 2001)

Further reading:

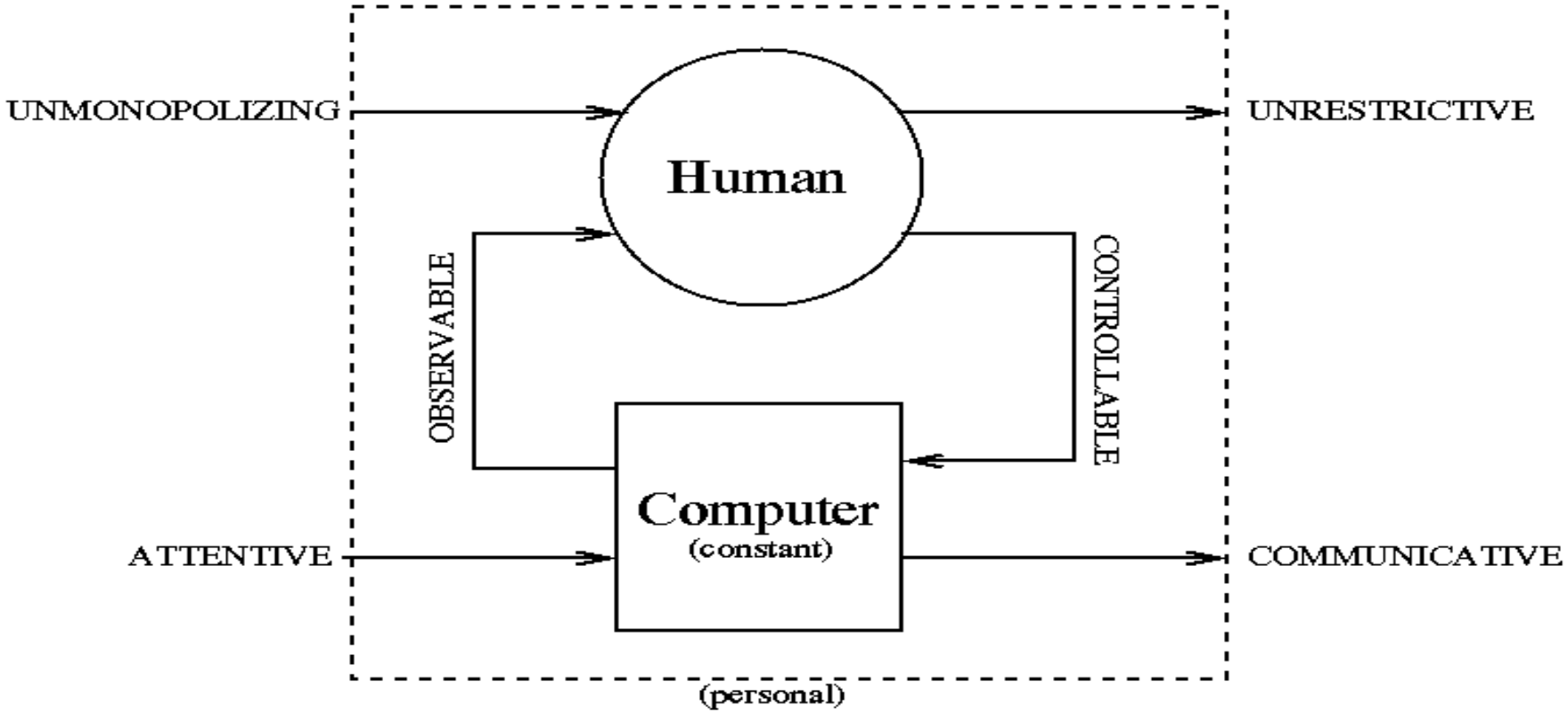
- Schilit et al., 1994
- http://en.wikipedia.org/wiki/Context_awareness

Context from smartphone



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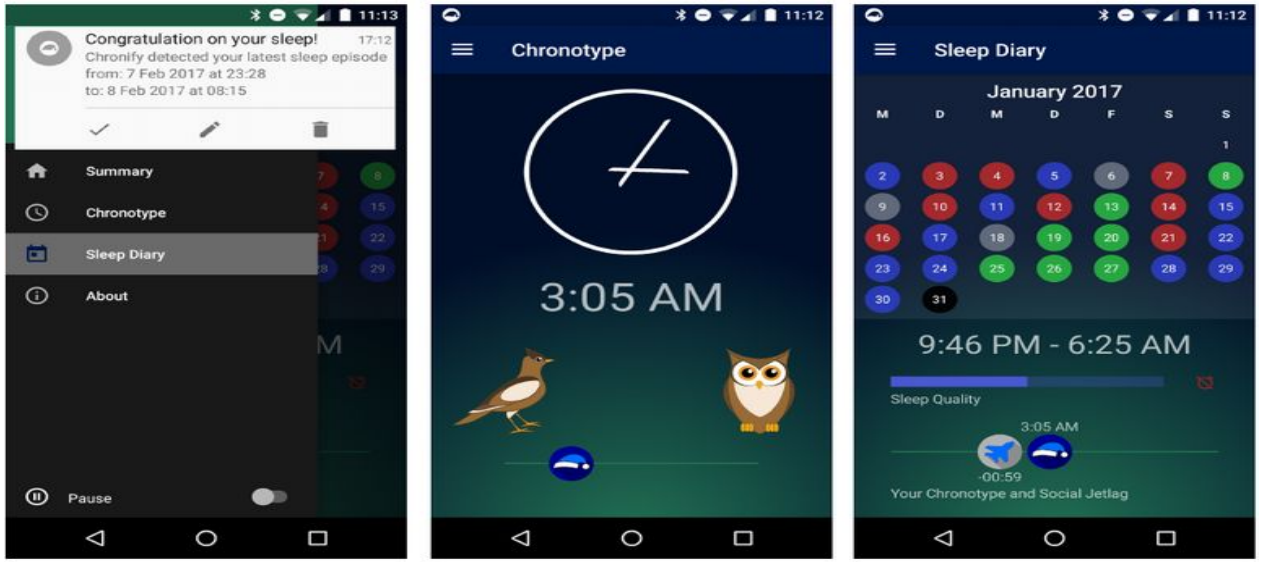
Ubiquitous, context-aware system properties



S. Mann: Definition of a Wearable Computer, ISWC 1998

Context-aware system (cont'd)

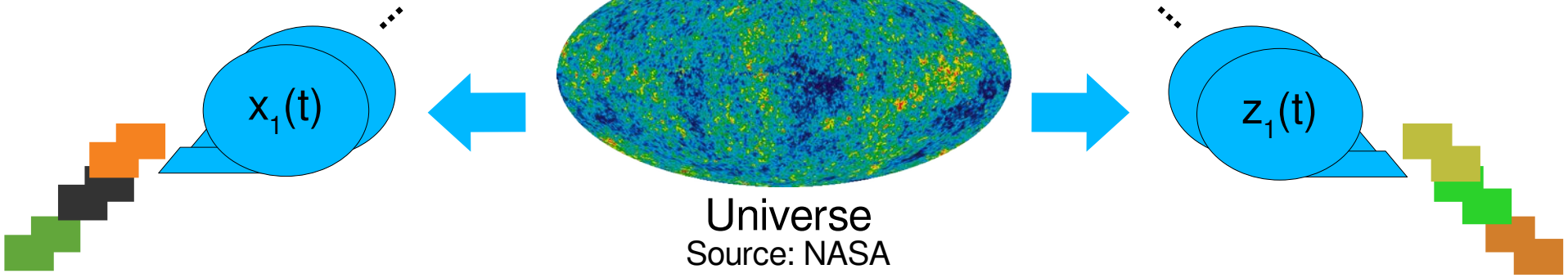
- **Unmonopolising / unrestrictive:** Do other things while using it.
- **Attentive / Communicative:** Monitor to the environment (augment perception).
- **Controllable / Observable:** Become part of the control loop; transparent operation.



Context Awareness: Theory & realisation

Prof. Dr. Oliver Amft

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Universe
Source: NASA

Random process of internal states:

$$\mathbf{X}(t) = \{ x_1(t), x_2(t), \ldots, x_N(t) \}$$

Problem 1

Only a fraction of the world is observable.

State-dependent process of <u>all potential</u> measurements:

$$\mathbf{Z}(t) = \{ z_1(t), z_2(t), \ldots, z_M(t) \} \triangleq F_{\text{Global}} [\mathbf{X}(t)]$$

Context: Very small observable (i.e. not hidden) subset of the internal process:

$$\mathbf{X'}(t) = \{ x_1(t), x_2(t), \ldots, x_{N'}(t) \}$$
where $N' \ll N$.

Problem 2

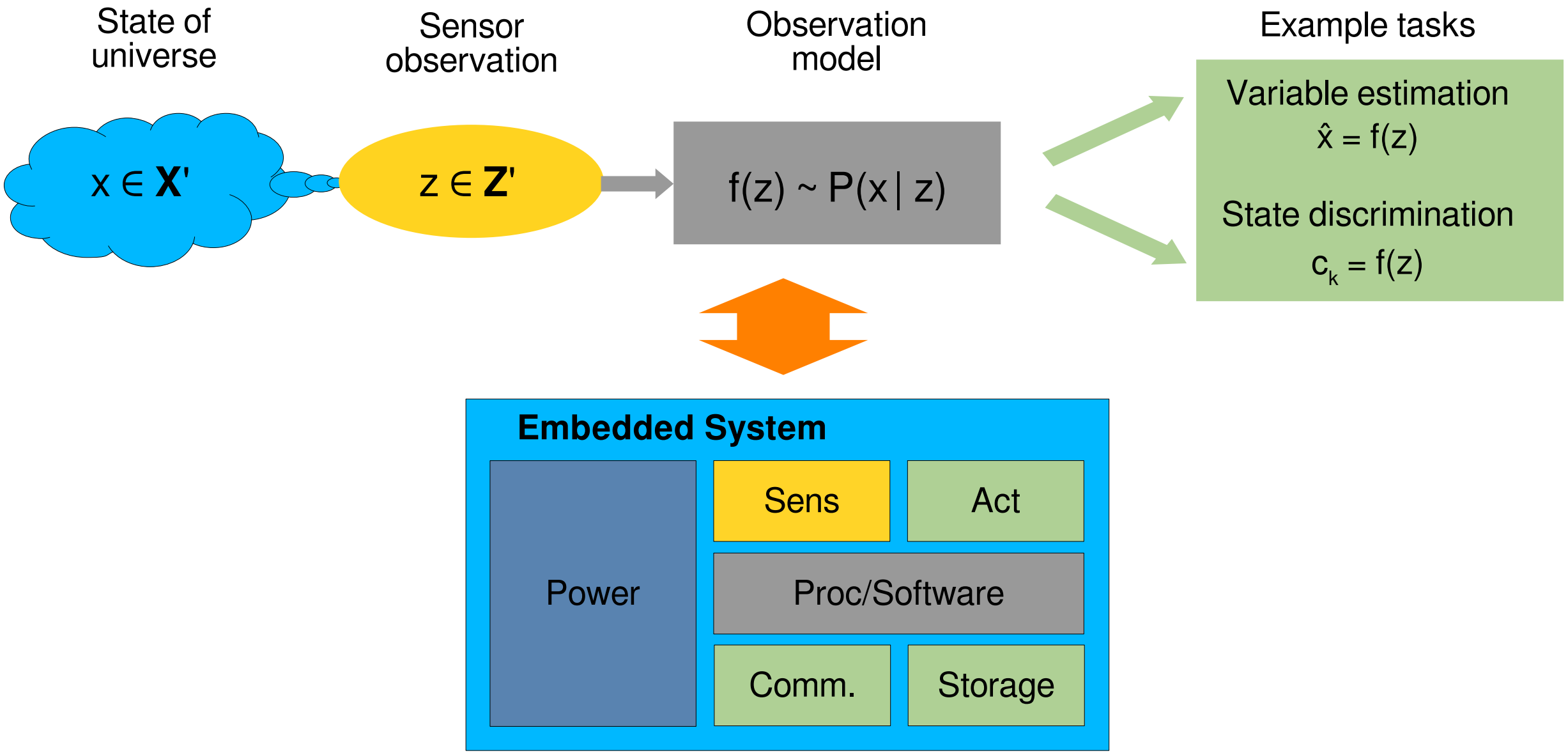
Measurements are sparse and noisy.

Measurement: Tiny subset of the possible observations:

$$\mathbf{Z'}(t) = \{ z_1(t), z_2(t), \ldots, z_{M'}(t) \}$$
where $M' \ll M$.

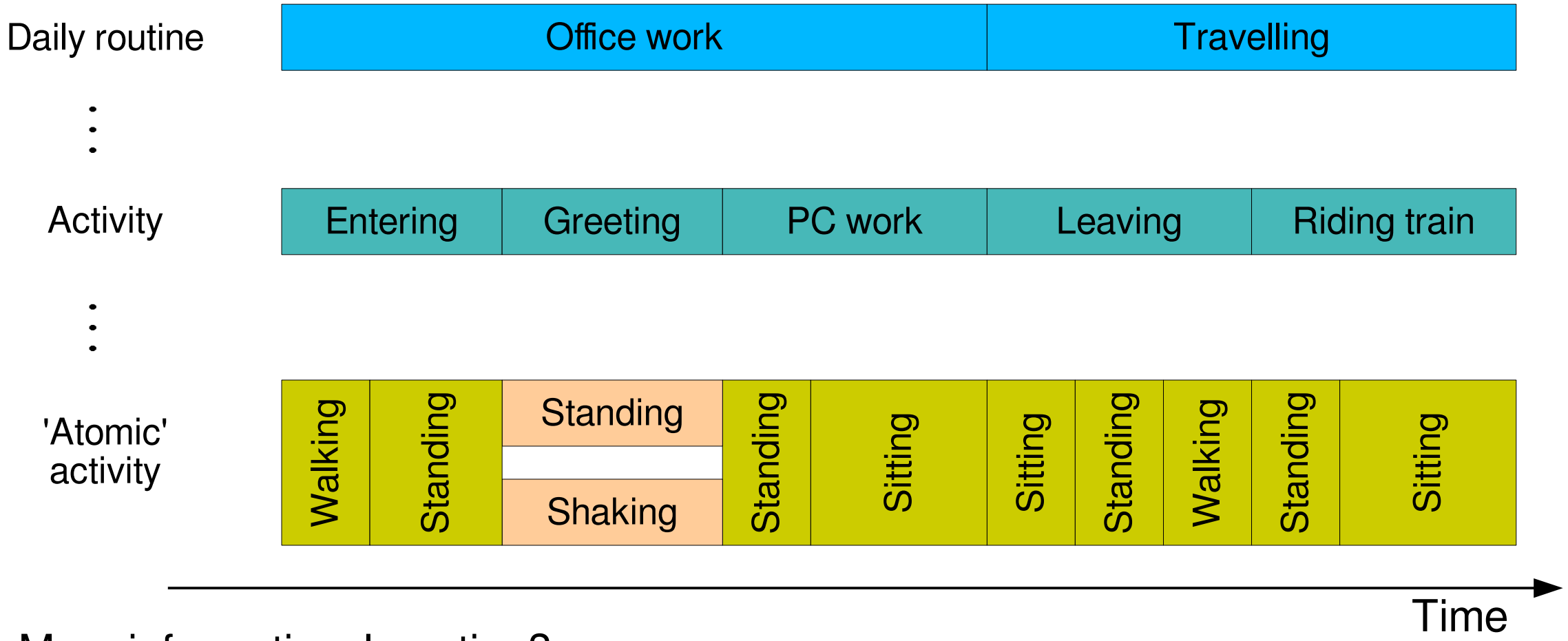
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Information model & System model



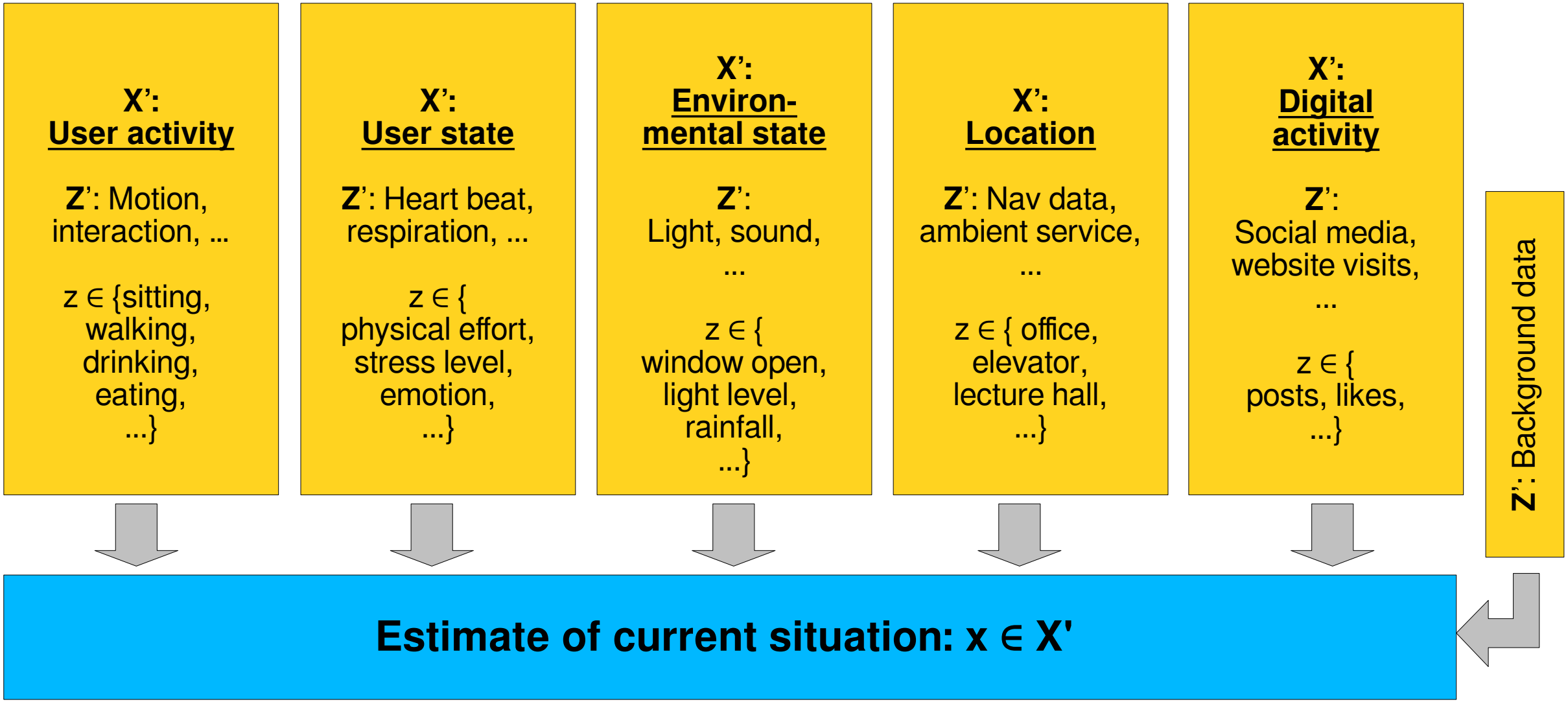
Context representation in time

Context can be represented in a temporal hierarchy:



More information: Location?

Context types



Context recognition examples



Fitness trackers



Sports skills assessment



Firefighter gear



Ward round information

Utilisation of context information

Active:
Application automatically adapts to discovered context, by changing the application's behaviour.



Passive:
Application presents the new or updated context to an interested user or makes context information persistent for the user to retrieve it later.



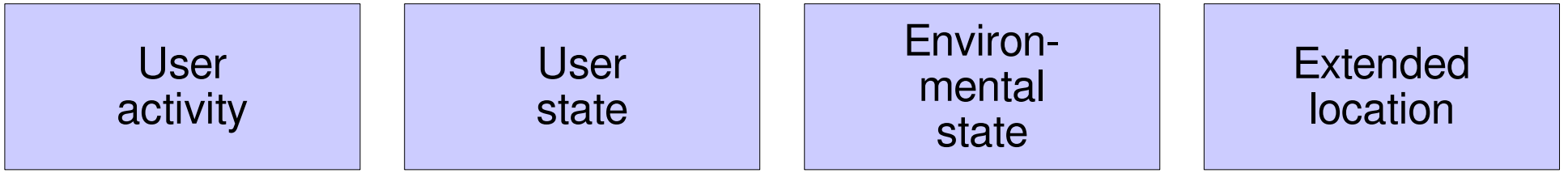
fitbit.com

Summary: context abstraction

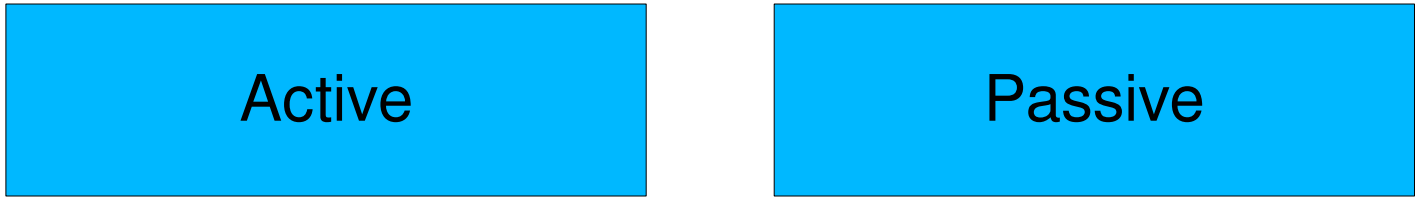
Context representation in time:



Context type representation:

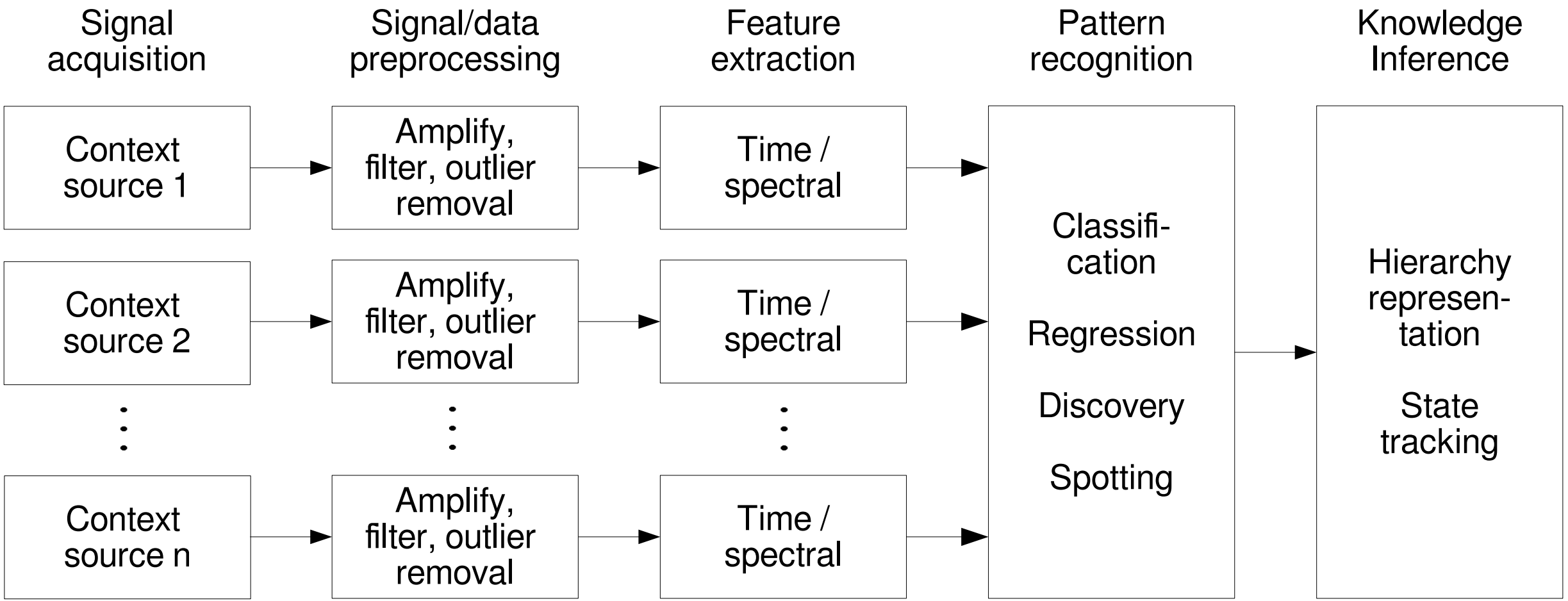


Modes of utilising context information:



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Context processing chain



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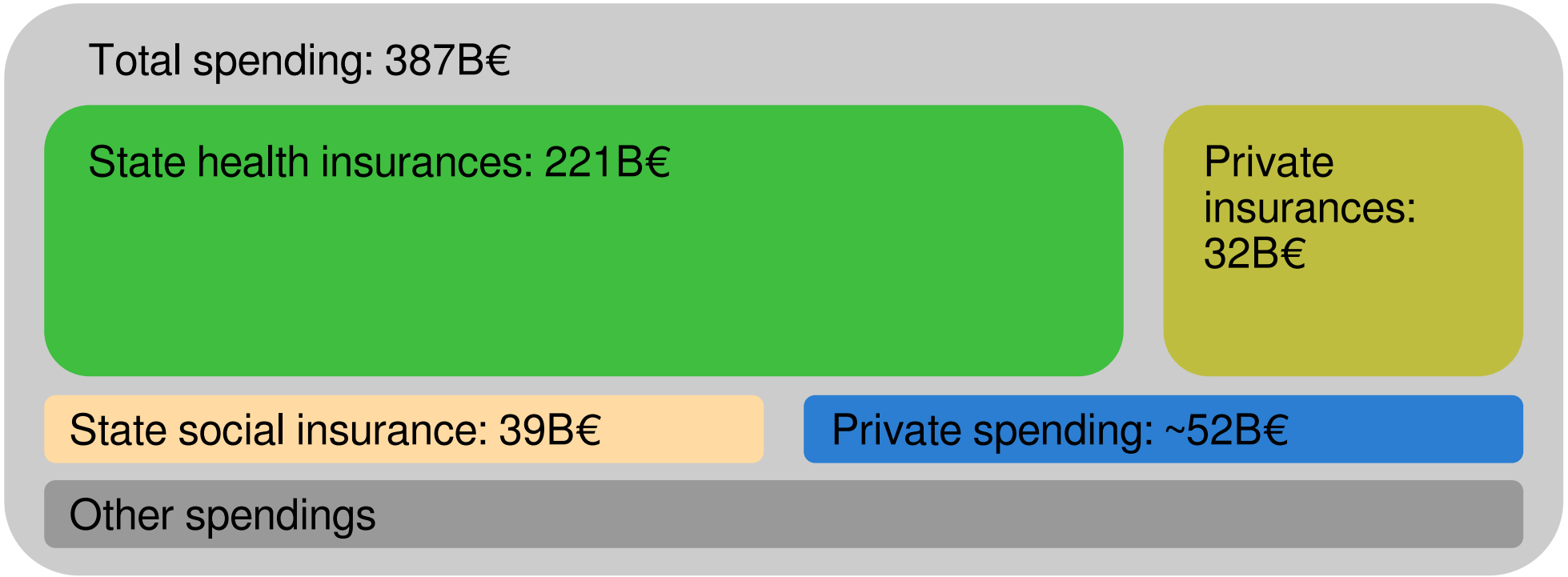
Digital Therapeutics (DTx) overview

Prof. Dr. Oliver Amft

amft@computer.org

Digital Health App market

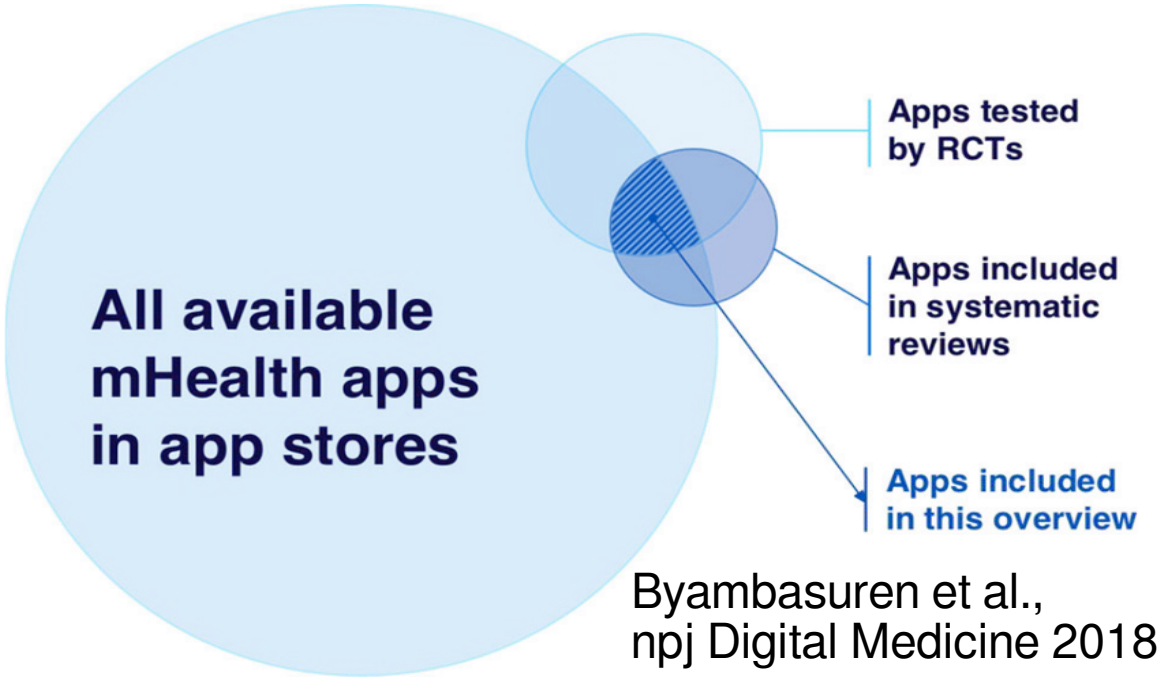
- Healthcare spendings in Germany (2018): 387B€
- Germany is the largest integrated healthcare market
- Digital Care Act (‘DVG’) provides regulation



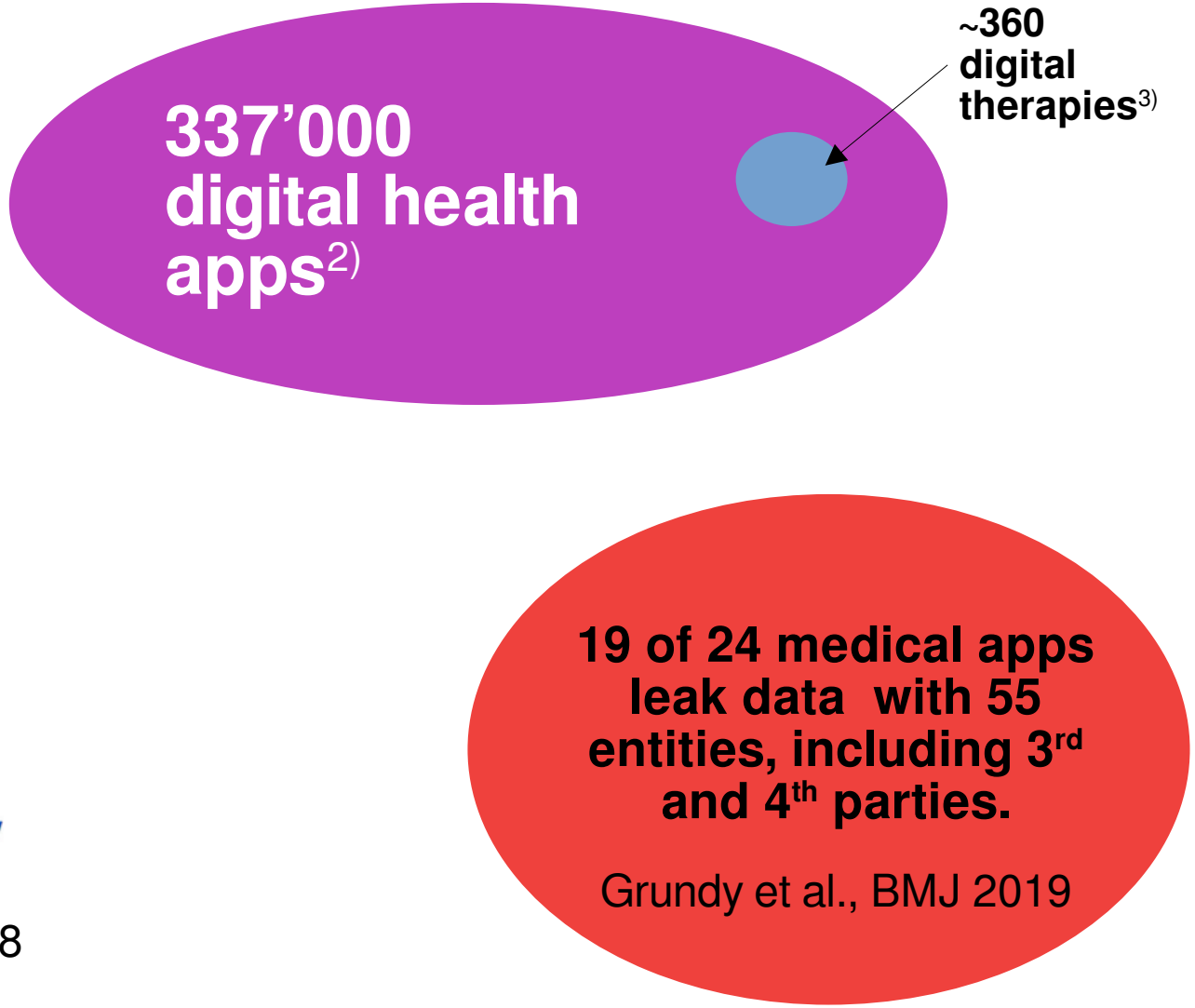
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Gesundheit/Gesundheitsausgaben/Tabellen/ausgabentraeger.html
https://www.vdek.com/presse/daten/d_versorgung_leistungsausgaben.html

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7.6 billion smartphones in 2025[1)]



Byambasuren et al., npj Digital Medicine 2018



[1)] Statista 2025, https://www.statista.com/statistics/330695/number-of-smartphone-users-worldwide
[2)] IQVIA Digital Health Trends 2024
[3)] IQVIA Institute for Human Data Science, 2021.

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Digital Therapeutics (DTx) challenges

34%

of 79 selected digital therapeutics had no published efficacy study[1)]

EVIDENCE

88%

of 20,991 mHealth apps contained code capable of collecting user data[2]

PRIVACY

38%

People with network coverage who do not use mobile internet; interoperability is an open problem[3)]

ADOPTION

[1)] Amyx et al., JAMA Intern Med 2024
[2)] Tangari et al., BMJ 2021
[3)] GSMA State of Mobile Internet Connectivity 2026; IQVIA Digital Health Trends 2025

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Digital Health App challenges

1) Evaluation of data privacy

2) Documenting positive care effects (continuously)

3) (Official) rating digital health apps

4) App market store and national revenue handling

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Digital Health
course organisation

Prof. Dr. Oliver Amft

amft@computer.org

Course programme, part 1

Course: Digital Health (DiHe), WS 2026/27
Lecturers: Prof. Oliver Amft, Dr. Mario Cypko, Dr. Göran Köber, Prof. Michael Quante, Prof. Heinz Lemke, Dr. Julius Wehrle, Dr. Lara Puhlmann, Dr. Isabel Nieto Alvarez
Exercises: Aditya Kumar
Zoom link: https://uni-freiburg.zoom-x.de/j/62400790245?pwd=Sf3ufktBw7i3ueOgGkrvE9ODtzL47P.1
Update: 16.10.2026

Date	Typ	Time	Location	Topics	Lecturer
19.10.2026	L1	12-14	HS 00 036 (G.-Köhler-Allee 101)	Introduction to Digital Health	Prof. Oliver Amft
22.10.2026	E1	16-18	Online	Machine Learning Refresher	Aditya Kumar
26.10.2026	L2	12-14	HS 00 036 (G.-Köhler-Allee 101)	Navigating Through Medical and Ambient Data: Convergence of Mobile Technologies, GDPR, and Contemporary Data Protection Strategies.	Dr. Mario Cypko
29.10.2026	E2	16-18	Online	Medical data: Introduction	Aditya Kumar
02.11.2026	L3	12-14	HS 00 036 (G.-Köhler-Allee 101)	Overview of AI-methods in health applications and medical decision support systems.	Dr. Mario Cypko
05.11.2026	E3	--	--	Medical data: Study week	Aditya Kumar
09.11.2026	L4	12-14	HS 00 036 (G.-Köhler-Allee 101)	Clinical Integration - Medical IT infrastructure with key standards, now and in the future	Dr. Julius Wehrle
12.11.2026	E4	--	--	Medical data: Study week	Aditya Kumar
16.11.2026	L5	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Biomarkers: Timeseries data	Dr. Göran Köber
19.11.2026	E5	16-18	Online	Medical data: Discussion and pitch	Dr. Mario Cypko?
23.11.2026	L6	12-14	Online	Digital Biomarkers: Mental Health	Dr. Lara Puhlmann
26.11.2026	E6	16-18	Online	Temporal data modelling: Introduction	Aditya Kumar

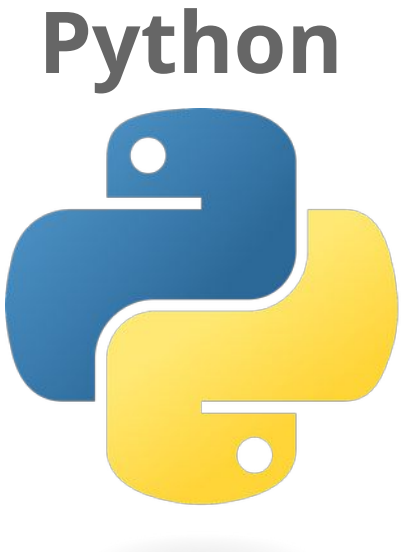
Course programme, part 2

30.11.2026	L7	12-14	HS 00 036 (G.-Köhler-Allee 101)	Human Digital Twin: Concept	Dr. Mario Cypko
03.12.2026	E7	--	--	Temporal data modelling: Study week	Aditya Kumar
07.12.2026	L8	12-14	HS 00 036 (G.-Köhler-Allee 101)	Modeling Digital Twins	Dr. Göran Köber
10.12.2026	E8	--	--	Temporal data modelling: Study week	Aditya Kumar
14.12.2026	L9	12-14	HS 00 036 (G.-Köhler-Allee 101)	Patient Experience Pathway	Dr. Isabel Nieto Alvarez
17.12.2026	E9	16-18	Online	Temporal data modelling: Discussion and pitch	Aditya Kumar
	--	--	--	--	--
11.01.2027	L10	12-14	HS 00 036 (G.-Köhler-Allee 101)	Population-scale digital twins	Dr. Göran Köber
	--	--	--	--	--
18.01.2027	L11	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Patient Twin & Model-Guided Medicine	Dr. Mario Cypko
21.01.2027	E10	16-18	Online	Bayesian Networks: Introduction	Dr. Mario Cypko
25.01.2027	L12	12-14	HS 00 036 (G.-Köhler-Allee 101)	Clinical Perspective on Digital Health	Prof. Michael Quante
28.01.2027	E11	--	--	Bayesian Networks: Study week	Dr. Mario Cypko
01.02.2027	L13	12-14	HS 00 036 (G.-Köhler-Allee 101)	Wisdom-oriented Healthcare System	Prof. Heinz Lemke
TBA	L14	16-18	TBA	Digital Health Research Night	Prof. Oliver Amft
04.02.2027	E12	16-18	Online	Bayesian Networks: Discussion	Dr. Mario Cypko
08.02.2027	L15	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Health applications: prescription models, regulations; General Q&A session	Prof. Oliver Amft

Meeting list is continuously updated.

In doubt, please consult the most recent version on ILIAS.

Exercise sessions



- Exercise sheets are provided one week in advance of the exercise discussion.
- Exercises goals: Practice implementing algorithms in Python and smartphone apps in Flutter.
- Exercise discussion: Information / Q&A on current exercise sheet.

- All exercise sessions take place online via Zoom.

- Each topic follows the same cycle:
  - **Introduction session:** the topic, the task notebook and the question sheet are introduced.
  - **Study week(s):** you work through the task and the sheet on your own and submit both by the stated deadline.
  - **Results session:** you pitch your results in a short presentation (about 5 minutes, not obligatory), and we discuss the solutions together.

- Solutions are uploaded after the results session (Python/Jupyter notebooks).

- Only students registered on ILIAS can take part in the exercises.

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Online materials

Slides

- Next lecture materials are available approx. one week in advance (if possible).
- Please answer all learning questions (quizzes) of the previous lectures before proceeding.

Learning questions (quizzes)

- Learning questions are provided on ILIAS to confirm comprehension, application, and generalisation of the concepts.

Forum

- Questions/insights shall be posted in the forum before the meeting.
- Open questions can be answered too. Make it your medium!

Other online materials

- Slides, exercise task descriptions, book(s).

Please connect to ILIAS

Link:
https://ilias.uni-freiburg.de/goto.php/grp/4384010

Password: digitalhealth2627

Deadline: 29.10.2026, 23:59

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Housekeeping notes

Participation & preparation

- Attend lectures and exercise sessions.
- Prepare for meetings using lecture/exercise materials + recommended literature + quizzes, course forum.
- Submit completed exercise notebooks.

Online materials

- All course material will be available on ILIAS.
- Slides, exercise task descriptions, book chapter(s).
- Discussion forum: questions, finding exercise group members.

Studienleistung (SL) & Exam

- Register for course on ILIAS by Oct 29, 2026.
- Written exam; tentative date: mid Feb 2027; questions include exercise + lecture material.
- Studienleistung (SL) (prerequisite for exam taking): Submit exercise and pass (code produces correct responses).

Please connect to ILIAS

Link:
https://ilias.uni-freiburg.de/goto.php/grp/4384010

Password: digitalhealth2627

Deadline: 29.10.2026, 23:59

Course contacts

For questions related to course content and general organisation, please use the **course forum** on ILIAS whenever possible.

Organisation:

- Emanuele Oppermann, e-mail: oppermann@informatik.uni-freiburg.de

Lectures:

- Dr. Mario Cypko, e-mail: Mario.Cypko@Hahn-Schickard.de
- Dr. Göran Köber, e-mail: koeberg@informatik.uni-freiburg.de
- Prof. Dr. Oliver Amft, e-mail: amft@cs.uni-freiburg.de
- Individual lecturers contacts provides in respective slide decks

Exercises:

- Aditya Kumar, e-mail: Aditya.Kumar@Hahn-Schickard.de

Please connect to ILIAS

Link:
https://ilias.uni-freiburg.de/goto.php/grp/4384010

Password: digitalhealth2627

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Discussion

Please connect to ILIAS

Link:
https://ilias.uni-freiburg.de/goto.php/grp/4384010

Password: digitalhealth2627

Deadline: 29.10.2026, 23:59

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Quiz

1) What sort of medical data types can mobile devices provide?
2) Why is the integration of mobile data clinically relevant?
3) What does the exposome include? What is the behaviourome?
4) Please describe an example of context awareness related to healthcare?
5) What context types can smartphones provide?
6) What is a digital health app according to the DiGA regulation?
7) Which requirements need to be fulfilled to enter an app into the DiGA Registry?
8) Describe the idea behind the self-certification and continuous certification?
9) Which basic elements lead from data to medical decision support within the scope of Digital Health?
10) How do doctors learn about digital health applications?
11) Can doctors prescribe an app?

Literature

- Fast Facts: Digital Medicine - Measurement. Coravos, A. et al. (S. Karger AG, 2020). doi:10.1159/isbn.978-3-318-06708-8.
- Wearables To Fight COVID-19: From symptom tracking to contact tracing. Amft, O., Lopera Gonzalez, L. I., Lukowicz, P., Bian, S. & Burggraf, P. IEEE Pervasive Computing 19, 53–60 (2020).
- How Wearable Computing Is Shaping Digital Health. Amft, O. IEEE Pervasive Computing 17, 92–98 (2018).
- What Will We Wear After Smartphones? Amft, O. and Van Laerhoven, K. Pervasive Computing, IEEE, 16(4), 2017.
- Context-aware computing. Dey, A.K. In: Ubiquitous Computing. Edited by John Krumm. Chapter 8, 2010.
- Machine learning in Action. Peter Harrington, Manning 2012 ISBN: 978-1617290183

